Computers and Education: Artificial Intelligence 9 (2025) 100474

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

Developing a theory-grounded AI tool for the generation of culturally responsive lesson plans

Matthew Nyaaba [a,b,c,*], Xiaoming Zhai [a,b,d]

[a] AI4STEM Education Center, University of Georgia, Athens, GA, USA
[b] National Center on Generative AI for Uplifting STEM+C Education (GENIUS Center), University of Georgia, Athens, GA, USA
[c] Department of Educational Theory and Practice, Mary Frances Early College of Education, University of Georgia, Athens, GA, USA
[d] Department of Mathematics, Science, and Social Studies Education, Mary Frances Early College of Education, University of Georgia, Athens, GA, USA

ARTICLE INFO

Keywords:
Custom GPT
Artificial intelligence (AI)
Generative AI
Automatic prompting
Theoretical prompt
Culture
Context
Prompt engineering

ABSTRACT

As educators begin using Generative AI (GenAI) for lesson planning, they often encounter generated content that fails to consider the classroom's cultural context. In this study, we address this issue by adopting a design science research approach to develop a theory-based prompt grounded in culturally responsive pedagogy (CRP) and using it to customize a *Culturally Responsive Lesson Planner (CRLP) GPT*. Guided by the CRP framework, the CRLP uses an *Interactive Semi-Automated* (ISA) prompt architecture that engages teachers in dialogue to collect cultural and contextual details before generating a lesson plan. To evaluate the CRLP's effectiveness, we asked two expert reviewers to compare Grade 7 "States of Matter" lesson plans for Ghana's Ashanti Region, generated with both the CRLP and the base GPT-4o using a standard prompt. The expert reviewers rated the CRLP-generated lesson plan higher in cultural elements identified (36 vs. 21 elements), accuracy (1.8 vs. 1.2), and curriculum relevance (2.0 vs. 1.3) than that created by the standard prompt within the base GPT-4o. The CRLP-generated lesson plan also included more Asante Twi examples such as "Solid" (*ɛpono* [wooden furniture], *dadeɛ* [metal objects], *aboɔ* [stones], and *ntadeɛ* [clothing]), recommended local teaching resources, and allowed teachers to make final revisions before generating the complete lesson plan. Additionally, the CRLP included the developer's contact details to encourage ongoing feedback and improvement. However, cultural hallucinations were slightly higher (0.75 vs. 0.5) in the CRLP-generated lesson plan compared with the standard GPT-4o prompt. These findings suggest that a GenAI tool grounded in educational theory is more effective in supporting the goals of education than the standard version. Furthermore, the CRLP and its ISA prompt strategy represent Human-in-the-loop system that has the potential to enhance teachers' AI literacy and culturally responsive pedagogy as they engage with the tool. We recommend future studies comparing CRLP and human-generated lesson plans, as well as empirical research that tests CRLP lesson plans in classroom settings.

1. Introduction

Generative AI (GenAI) has emerged as a transformative educational tool with significant potential to enhance teaching and learning experiences (Collins et al., 2024; Shalong et al., 2024). However, research consistently demonstrates that GenAI models, typically trained on datasets dominated by Western perspectives, often produce culturally insensitive outputs that fail to reflect diverse cultural contexts (Watson et al., 2024; Yogarajan et al., 2024). This limitation is particularly problematic in educational settings where culturally responsive pedagogy is essential for equitable learning. For instance, when asked to list the four seasons, GenAI tools frequently default to Western seasonal patterns, disregarding other cultural perspectives, exemplifying the pervasive Western bias that can shape learners' perceptions and potentially hinder their understanding of diverse global viewpoints (Nyaaba et al., 2024).

The need for culturally responsive educational resources is especially critical in education across the world and particularly in developing countries and diverse educational settings (Shahjahan et al., 2022), where reliance on Western curricula and English-dominated instruction often marginalizes local cultures and languages (Bautista et al., 2021; UNESCO, 2023a). Although educators increasingly use GenAI for lesson planning, they continue to raise concerns about inaccuracies and the lack of cultural and contextual relevance in generated outputs

* Corresponding author. AI4STEM Education Center, University of Georgia, Athens, GA, USA.
E-mail addresses: matthew.nyaaba@uga.edu (M. Nyaaba), xiaoming.zhai@uga.edu (X. Zhai).

https://doi.org/10.1016/j.caeai.2025.100474
Received 26 December 2024; Received in revised form 28 August 2025; Accepted 5 September 2025
Available online 11 September 2025
2666-920X/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

(Karpouzis, 2024; Lee & Zhai, 2024). Customized Generative Pre-trained Transformer (GPT) offer the potential for improvement, yet their effectiveness remains heavily dependent on prompt engineering skills that many educators lack, particularly in underrepresented regions (Meincke et al., 2024; Yuan & Hu, 2024). Gizzi and Felici (2024) indicated that effective prompt engineering in educational contexts requires specialized knowledge to guide AI outputs toward contents that are accurate, contextually relevant, and culturally appropriate (Garrido-Merchán et al., 2024).

This study addresses this gap by developing a methodological approach for creating culturally responsive custom GPTs through an *Interactive Semi-Automated* (ISA) prompt architecture, which removes prompt engineering as a barrier for educators without technical expertise. Our approach is primarily grounded in Nyaaba et al.'s (2024) GenAI-Culturally Responsive Science Assessment (GenAI-CRSciA) framework, which identifies five essential dimensions for culturally responsive GenAI applications: *Indigenous Language Integration, Indigenous Knowledge Systems, Religion/Spirituality, Race/Ethnicity, and Family and Community Engagement*. We enhance this foundation with elements from Gay's (2018) *Culturally Responsive Teaching framework* and Ladson-Billings' (1994) *Culturally Relevant Pedagogy* to create a theoretically grounded prompt structure that systematically guides users in providing contextual parameters to generate culturally responsive content.

The methodological process employed a design science research, integrating these theoretical frameworks into a structured prompt architecture implemented within a custom GPT, *Culturally Responsive Lesson Planner* (CRLP) designed to generate lesson plans across disciplines. We first developed theoretically grounded prompts, scripted them in Python with the OpenAI API and tested and refined the cultural parameter collection and integration strategies (Chukhlomin, 2024; Rouzegar & Makrehchi, 2024). This technical development phase allowed precise control over prompt structures and cultural responsiveness in generated outputs. Recognizing that this technical approach would remain inaccessible to most educators, we translated our validated prompt architecture into a custom GPT implementation with the ISA prompt system, the CRLP that eliminates technical barriers while maintaining the theoretical integrity. Through a comparative evaluation of lesson plans generated by the CRLP versus the standard prompt approach within base GPT-4o, we demonstrate that GenAI can support teachers in developing culturally responsive lesson plans that reflect learners' diverse cultural contexts and enhance learning. The following questions guided the study:

1. What cultural elements emerge in GenAI-generated lesson plans when educators use a theory-informed ISA prompt architecture in a custom GPT?
2. How do lesson plans generated through a theory-informed ISA prompt approach in a custom GPT differ from those generated by standard prompt in a base GPT in addressing cultural responsiveness?

2. Literature review

2.1. Recent studies on OpenAI customized GPTs in education

The application of GenAI through customized GPT models is rapidly advancing in educational contexts (Sathe et al., 2024). Research has demonstrated the significant potential of customized GPTs to enhance student engagement, support subject-specific learning, and improve assessment practices (Collins et al., 2024; Shalong et al., 2024). Rouzegar and Makrehchi (2024) compared GPT-3.5 and GPT-4 in generating customized test questions aligned with active learning principles; they found that GPT-4 produced more nuanced, contextually relevant questions that better aligned with course objectives and supported critical thinking. Their research emphasizes the importance of prompt engineering training for educators to maximize question quality and alignment with educational goals, highlighting how technical expertise mediates the effectiveness of these tools (Rouzegar & Makrehchi, 2024).

Customized GPT models have demonstrated effectiveness in subject-specific applications. Collins et al. (2024) introduced AnatomyGPT, a customized model offering interactive quizzes, detailed explanations, and specialized insights tailored to anatomical sciences students. Their iterative development process, incorporating student feedback to improve accuracy and relevance, underscores the importance of collaboration between educators and GenAI developers in creating models that remain pedagogically aligned and responsive to specific disciplinary needs (Collins et al., 2024). Similarly, Chukhlomin (2024) explored custom GPTs for strategic planning in higher education, finding that they effectively generate insights aligned with institutional goals while emphasizing the critical role of human expertise in interpretation and implementation.

Empirical research has begun to document the measurable educational benefits of customized GPTs. Shalong et al. (2024) conducted a 14-week randomized controlled trial evaluating LearnGuide, a custom ChatGPT tool for medical education. Their findings demonstrate significantly higher Self-Directed Learning Scale scores, Cornell Critical Thinking Test scores, and Global Flow scores among students using the AI-supported system compared to students in the control condition, providing quantitative evidence of AI tools' potential to enhance educational outcomes (Shalong et al., 2024). These documented benefits have encouraged institutional adoption, with OpenAI developing ChatGPT Edu specifically for academic institutions based on successful implementations by individual professors at leading universities (OpenAI, 2024a; OpenAI, 2024b).

Notable customized GPT applications demonstrate the diversity of educational implementations. Professor Nabila El-Bassel at Columbia University leveraged a custom GPT to process community health data for public health interventions, while Professor Ethan Mollick at the Wharton School employed customized GPTs to foster critical thinking in reflective assignments (OpenAI, 2024a; OpenAI, 2024b). Professor Christiane Reves at Arizona State University developed "Language Buddy," a GPT that adapts to German learners' proficiency levels, supporting language acquisition while reducing faculty assessment time (King, 2024). Most relevant to this current study is the work of Nyaaba et al. (2024) who developed the CRSciA-Generator, a customized GPT designed to generate culturally responsive science assessment items, and whose methodological framework provides the foundation for this study.

2.2. Concerns and limitations of generative AI in education

With the numerous potential benefits, GenAI applications in education face significant challenges, particularly regarding cultural bias and privacy concerns. GenAI models, typically trained on datasets dominated by Western perspectives, often produce culturally insensitive outputs that fail to reflect diverse cultural contexts (Watson et al., 2024; Yogarajan et al., 2024). This Western-centric bias is especially problematic in educational settings, where it can shape learners' perceptions and potentially hinder their understanding of diverse global viewpoints (Godwin-Jones, 2024; Baker & Hawn, 2022). Research by Ali et al. (2024) indicates that Gen Z users, a key demographic in digital education, are particularly influenced by these Western-dominated outputs, while Watson et al. (2024) demonstrate that fine-tuning AI with region-specific data can substantially improve cultural relevance and

enable more meaningful engagement in non-Western contexts.

Beyond cultural bias, GenAI technologies raise important privacy concerns due to their use of personal data for training and output generation. Researchers highlight multidisciplinary ethical challenges, including data protection issues, and they advocate for proactive, policy-driven approaches to responsible implementation (Al-Kfairy et al., 2024; Xiang, 2024). These combined challenges underscore the need for culturally responsive, privacy-conscious approaches to GenAI implementation in educational settings, particularly when serving diverse and underrepresented populations.

2.3. Mitigating GenAI concerns

Researchers and policymakers have proposed several strategies to address the cultural bias and privacy concerns associated with GenAI in education (Pescapè, 2024). To mitigate cultural limitations, studies advocate for personalized fine-tuning frameworks that integrate local values and narratives, enabling GenAI systems to serve diverse cultural contexts more effectively (Ali et al., 2024; Watson et al., 2024). This technical approach aligns with policy recommendations from Abdulrazaq (2024), who suggests that governments should adopt GenAI models specifically tailored to local cultural standards and establishing country-specific data standards. At the regulatory level, the European Union's proposed GenAI Act aims to enhance transparency by mandating disclosures about the logic and data used by GenAI systems (Justo-Hanani, 2022), while other frameworks recommend documentation requirements and certifications for high-risk GenAI applications, such as educational tools (Delipetrev & Gómez, 2023; Zhai & Krajcik, 2022).

Regarding privacy protection, platforms like OpenAI have implemented user rights frameworks allowing individuals to access, update, delete, or transfer personal data, as well as restrict its processing and withdraw consent (OpenAI, 2024a; OpenAI, 2024b). Among these approaches, custom GPTs have emerged as an especially promising solution, offering educators the ability to create purpose-built applications that address specific cultural and contextual needs. As demonstrated by Collins et al. (2024), custom GPTs like AnatomyGPT can incorporate iterative feedback from users to improve cultural relevance and accuracy, providing a practical pathway for developing more inclusive and contextually appropriate educational AI tools in diverse settings.

2.4. Prompt engineering for teaching and learning

Prompt engineering has become an essential tool in educational contexts, particularly as generative AI models like large language models (LLMs) gain prominence in enhancing learning experiences (Meskó, 2023). Through the thoughtful design of prompts, educators can guide AI outputs to be contextually accurate, relevant, and aligned with specific educational objectives. Gizzi and Felici (2024) illustrate this point in STEM education, demonstrating that well-crafted prompts in human movement sciences can help students understand complex concepts by customizing AI responses to meet their learning needs. Their study underscores that prompt engineering can play a significant role in STEM, where subject matter complexity requires precise explanations. By refining prompts to focus on particular content areas or scientific principles, educators can facilitate deeper engagement and comprehension, making difficult subjects more accessible (Gizzi & Felici, 2024).

In collaborative and project-based learning settings, prompt engineering also enhances peer-to-peer interactions by fostering higher-quality feedback. For instance, Sajadi et al. (2024) explore the role of prompts in improving peer evaluations within project-based learning contexts. They show that structured prompts enable generative AI to offer constructive feedback, which supports students in critically assessing one another's work and provides a scaffolded learning experience. This approach aligns GenAI output with pedagogical goals, emphasizing critical thinking and feedback literacy as essential skills in modern education. Moreover, Yuan and Hu (2024) examine prompt engineering in the context of higher education course evaluations, finding that specific, structured prompts allow GenAI models to generate meaningful and relevant feedback on course content and teaching quality. Their findings suggest that carefully designed prompts can shape LLM responses to focus on critical elements of the student experience, contributing to a more insightful and valuable evaluation process for instructors (Yuan & Hu, 2024; Joubin, 2024).

Furthermore, training students in prompt engineering has shown promise for optimizing GenAI-assisted learning and enhancing digital literacy (Knoth et al., 2024). Handa, Pal, and Ambasana (2023, November 24) discuss a program in which students were taught prompt engineering techniques to interact more effectively with GenAI chatbots. The study found that prompt engineering not only improved the quality of chatbot responses but also equipped students with the skills needed to navigate GenAI-based educational tools independently. Additionally, Viveros-Muñoz et al. (2024) highlight the impact of prompt structure on GenAI responses in multilingual settings, emphasizing the importance of culturally and linguistically aware prompt design. Their work reveals that prompt engineering can influence not just accuracy but also the cultural sensitivity of AI-generated content, which is vital in diverse educational environments. These studies underscore that prompt engineering is more than a technical skill; it is a crucial aspect of teaching that ensures that GenAI models serve as effective, inclusive, and adaptable educational resources.

2.5. Culturally responsive frameworks for GPTs

Culturally responsive education aims to create inclusive, equitable learning experiences by integrating diverse cultural values, experiences, and perspectives into teaching practices and curricula (Chen et al., 2022; Muñiz, 2019). This approach enhances student engagement and learning outcomes by recognizing cultural assets, but implementation faces significant challenges, particularly in contexts marked by systemic inequities and limited resources (UNESCO, 2023a; Esteban-Guitart and Moll, 2014). Teachers in underserved regions often lack access to necessary training and culturally diverse materials, especially in developing countries where an overreliance on Western curricula and English-medium instruction marginalizes local languages and knowledge systems (Nyaaba et al., 2024; UNESCO, 2020). GenAI technologies offer promising opportunities to address these barriers by enabling culturally responsive education on a large scale, potentially assisting teachers in creating localized resources, generating culturally relevant lesson plans, and providing multilingual support for diverse communities (UNESCO, 2023b).

Three theoretical frameworks provide essential foundations for developing culturally responsive GenAI applications. Gay's (2018) Culturally Responsive Teaching framework emphasizes cultural competence, academic success, and critical consciousness, guiding the design of GenAI prompts that reflect diverse backgrounds while maintaining high standards. Ladson-Billings' (1994) Culturally Relevant Pedagogy contributes principles of student engagement, cultural integrity, and empowerment, highlighting the importance of creating GenAI tools that honor students' identities and lived experiences (Kashyap et al., 2024). Most directly applicable, Nyaaba et al.'s (2024) GenAI-CRSciA framework identifies five essential dimensions for culturally responsive AI: Indigenous language, Indigenous knowledge,

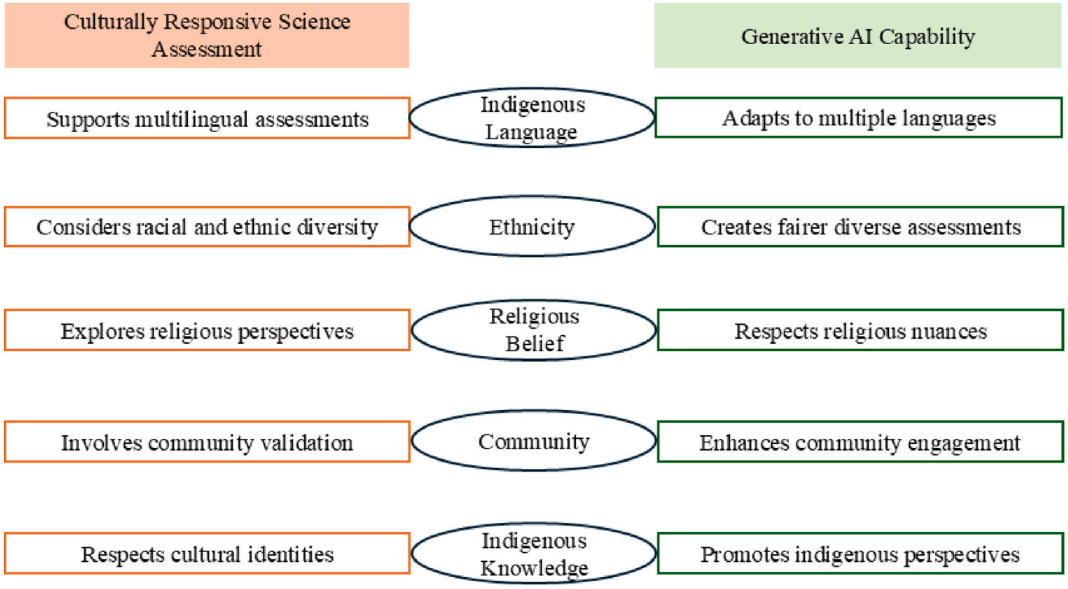


Fig. 1. Generative AI culturally responsive science assessment framework (adopted from Nyaaba et al., 2024)

religion, race/ethnicity, and family–community engagement (see Fig. 1). This framework provides a structured approach for developing GPTs that move beyond generic outputs to deliver content authentically reflecting students' backgrounds, addressing educational disparities by making learning more accessible and personalized for diverse learners.

Together, these frameworks establish theoretical foundations for developing GenAI applications that can effectively support culturally responsive education across diverse contexts.

3. Method

We adopted design science research (DSR) to address the persistent problem that generic GenAI lesson plans overlook learners' cultural and linguistic contexts (Peffers et al., 2007). DSR's core elements include: problem identification and objectives; theory-grounded artifact design and development; demonstration in context; rigorous evaluation of utility, quality, and efficacy; and communication of generalizable design knowledge, which guided this study (Gau et al., 2025; Peffers et al., 2007). Following this, our CRLP was designed and evaluated via an ISA prompt schema grounded in Culturally Responsive Pedagogy, Funds of Knowledge, and bilingual/translanguaging pedagogy. The ISA incorporates non-optional cultural "slots" (local language, community assets, representation, low/no-cost materials) across objectives, activities, and assessment. Development proceeded in two phases: first, a Python/OpenAI API prototype for precise control and internal validation, then a custom GPT for educator accessibility. This was followed by demonstrations that generate fully localized science lessons and an evaluation combining expert rubric ratings for CRP alignment and scientific accuracy.

3.1. Establishing theoretical-based prompt

Our prompt development centered on Nyaaba et al.'s (2024) GenAI-CRSciA framework, which provides a specialized structure for culturally responsive AI applications through five essential dimensions: *Indigenous Language Integration, Indigenous Knowledge Systems, Religion/Spirituality, Race/Ethnicity, and Family–Community Engagement*. We chose this framework specifically for its direct applicability to AI-generated educational content, providing clear parameters for embedding cultural responsiveness into prompt design. We enriched this foundation with complementary elements from Gay's (2018) principles of validating cultural identities and comprehensive integration and Ladson-Billings' (1994) dimensions of academic success through cultural competence and critical consciousness.

This theoretical synthesis yielded an integrated Culture and Context-Aware Framework (Fig. 1) that guided our ISA prompt architecture. The architecture translates these abstract theoretical principles into operational prompts through a structured sequence of questions that systematically collect cultural parameters across each dimension. For example, the prompt explicitly gathers information about regional language use, cultural practices, and community contexts, then instructs the model to integrate these elements throughout all components of the lesson plan rather than as isolated features. This approach ensures that cultural responsiveness is embedded throughout the instructional design (from objectives and vocabulary to activities and assessment), creating a cohesive, culturally grounded learning experience that reflects students' lived realities while maintaining academic rigor. By grounding our prompt design in established theoretical frameworks, particularly the GenAI-CRSciA model specifically developed for AI applications, we created a theoretically sound, practically implementable system for generating educational content that authentically reflects diverse cultural contexts.

3.2. Dual development approach

Our development process employed a dual approach, beginning with Python-based prompt engineering using the GPT-4o API and then transitioning to a custom GPT implementation within the ChatGPT platform to increase accessibility for educators. This dual development approach allowed us to first validate our theoretical approach using the precision of Python programming, then make it accessible to educators through the more user-friendly custom GPT interface, ultimately creating a practical tool for generating culturally responsive educational materials.

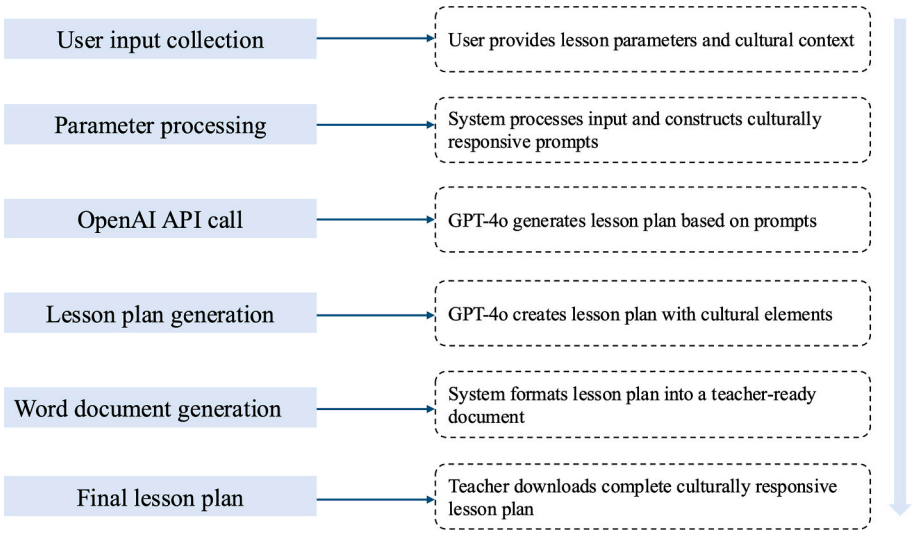


Fig. 2. The OpenAI Python script lesson plan generation workflow.

3.2.1. Python API implementation

The Python implementation followed a structured six-step workflow, as illustrated in Fig. 2. The process begins with user input collection (Step 1), where educators provide lesson parameters and optional cultural context information, preserving teacher agency throughout. The system then processes these inputs to construct theoretically grounded, culturally responsive prompts incorporating the five dimensions from our integrated framework (Step 2). These prompts are submitted to the OpenAI API (Step 3), which leverages GPT-4o's knowledge base to generate comprehensive lesson plans with embedded cultural elements (Step 4). The system automatically formats the output into teacher-ready documents in both plain text and Word formats (Step 5), ensuring practical usability in diverse educational settings. Finally, the teacher receives a complete, culturally responsive lesson plan ready for classroom implementation or further customization (Step 6) [Access the link to the Python script in Appendix A].

This workflow demonstrates how our theoretical framework was operationalized through a practical, end-to-end system that combines AI capabilities with educator expertise to produce culturally responsive educational materials. This workflow provides a methodical sequence, transforming cultural and educational parameters into a complete, ready-to-use lesson plan document.

3.2.1.1. Python environment setup. We established the Python environment to interact with the OpenAI API. This setup included importing essential libraries for API interaction, prompt processing, and result output. The configuration parameters were set as follows: the temperature was set at 0.7 to balance creative adaptation with factual consistency; token limits were established at 3000 to accommodate comprehensive lesson plans; and nucleus sampling (*top_p*) was configured at 0.95 to maintain natural language while reducing hallucinations (see Box 1). The system role message establishes the AI's persona as an educational expert specifically focused on culturally responsive approaches.

```
import openai
from docx import Document
from IPython.display import FileLink, display
# Set your API key - remember to secure this in a production environment
client = openai.OpenAI(api_key="YOUR_API_KEY")  # Replace with your API key
# Configuration parameters for optimal performance
CONFIG = {
    "model": "gpt-4o",        # Most advanced model for cultural nuance
    "temperature": 0.7,       # Balance of creativity and consistency
    "max_tokens": 3000,       # Sufficient for comprehensive lesson plans
    "system_role": "You are an expert curriculum developer specializing in culturally responsive education."
}
```

Box 1. Model Configuration

3.2.1.2. Parameter and GPT knowledge utilization. The parameter collection phase was structured to capture the multidimensional aspects of cultural and educational contexts necessary for culturally responsive lesson planning. Our approach leveraged GPT-4o's existing knowledge of diverse cultures and regions. The system was designed to collect specific contextual parameters from educators and then guide GPT-4o to access and apply its pre-trained knowledge to those cultural contexts (see Box 2).

3.2.1.3. Integrated prompt structure with cultural frameworks. Following our theoretical grounding, we developed a Python-based prompt generation function that explicitly operationalizes the five dimensions of cultural responsiveness while maintaining the flexibility to adapt to

```
Python
# Function to get user input for cultural and educational context
def get_user_input():
    print("=== Culturally Responsive Lesson Plan Generator ===")
    print("Please provide the following information:")
    # Educational context parameters
    country = input("Country (e.g., USA, Kenya, Ghana): ")
    region = input("Region (e.g., Ashanti): ")
    grade = input("Grade level (e.g., 7): ")
    school_level = input("School level (e.g., Junior High School): ")
    # Subject parameters
    subject = input("Subject (e.g., Science): ")
    topic = input("Topic (e.g., Matter): ")
    learning_objectives = input("Learning objectives: ")
    # Classroom parameters
    class_size = input("Class size (e.g., 30): ")
    lesson_duration = input("Lesson duration in minutes (e.g., 50): ")
```

```
# Optional cultural parameters to supplement GPT-4o's knowledge
    dominant_language = input("Dominant local language (optional, press Enter to let AI determine): ")
    cultural_notes = input("Any specific cultural notes you'd like to include (optional): ")
    # Create a parameters dictionary with user input
    parameters = {
        "country": country,
        "region": region,
        "grade": grade,
        "school_level": school_level,
        "subject": subject,
        "topic": topic,
        "class_size": class_size,
        "lesson_duration": lesson_duration,
        "learning_objectives": learning_objectives
    }
    # Only add optional parameters if provided
    if dominant_language:
        parameters["dominant_language"] = dominant_language
    if cultural_notes:
parameters["cultural_notes"] = cultural_notes

    return parameters
```

Box 2. Context Input

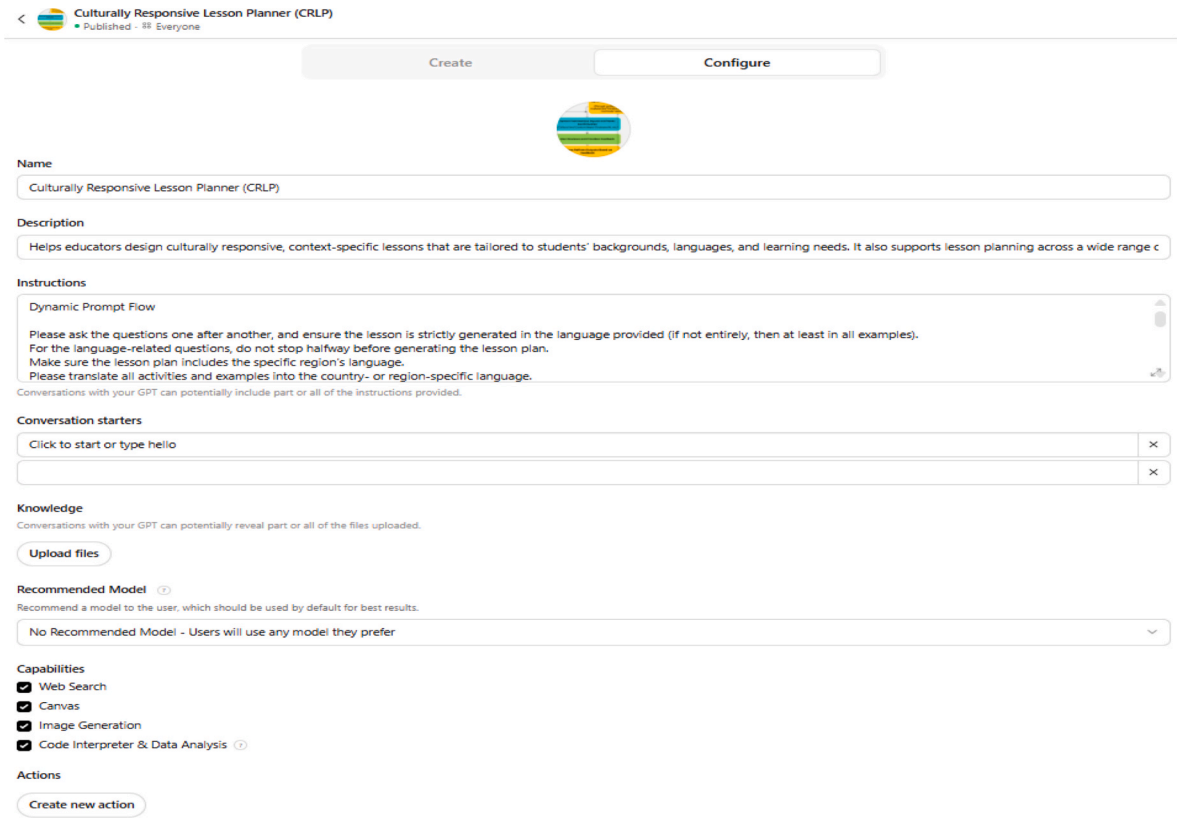


Fig. 3. Configuration page of the culturally responsive lesson planner.

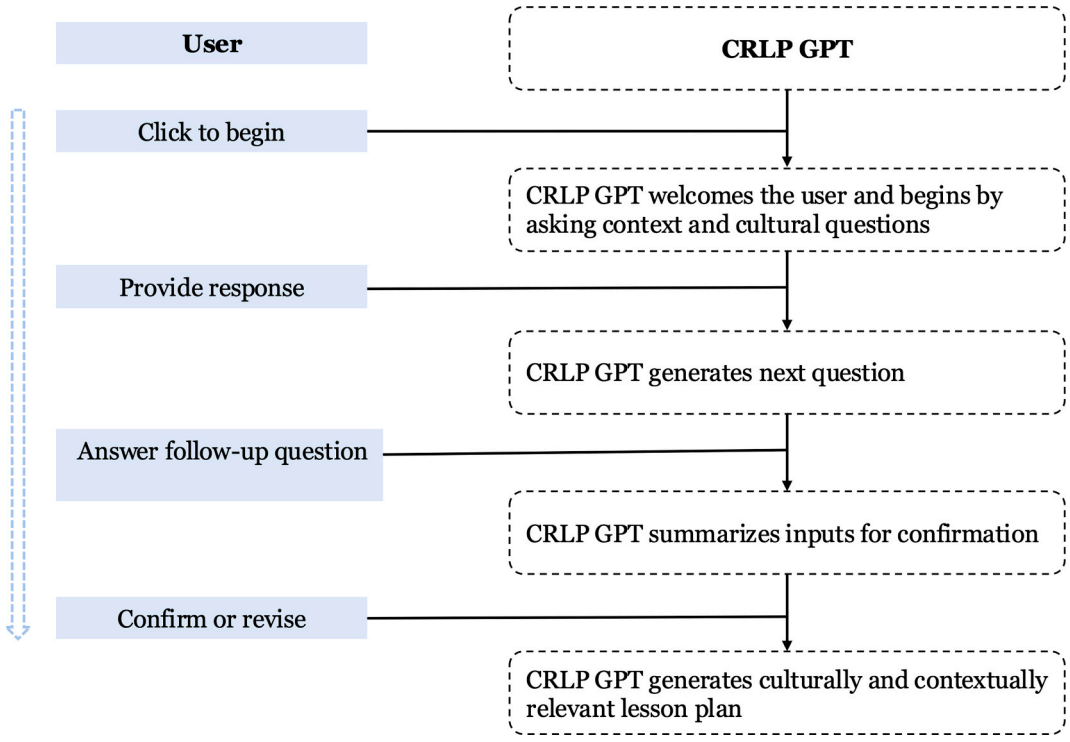


Fig. 4. The interactive semi-automated prompt for culture and context-aware GPTs (adapted from Nyaaba et al., 2024).
NB: *Culturally Responsive Lesson Planner (CRLP) — Custom Generative Pre-trained Transformer (GPT)*

Table 1
Difference between interactive semi-automated prompt system and the standard prompt.

Criteria	Interactive Semi-Automated Prompt System	Standard OpenAI GPT Prompt
Prompt Interaction	Dynamic, iterative	Static, one-time
Cultural Context Customization	High (guided, detailed)	Moderate (user-dependent)
User Engagement & Validation	Extensive	Minimal
Complexity & Usability	Higher complexity (multiple interactions)	Lower complexity (simpler)

diverse cultural contexts. The core implementation of this approach is reflected in the generate_full_prompt() function, which constructs a theoretically grounded prompt based on user-provided parameters such as country, region, grade level, subject, and topic (see Box 3). This function systematically integrates educational context parameters with explicit instructions addressing each dimension of cultural responsiveness (see Box 4). The structure ensures that cultural elements are not merely added as separate components but are thoroughly integrated throughout the entire lesson plan, from objectives and materials to activities and assessments.

Table 3
Rating scale definitions for cultural element evaluation.

Rating Dimension	Score	Definition
Accuracy	0	Cultural reference is incorrect or misrepresented
	1	Partially accurate with minor errors or oversimplifications
	2	Fully cultural reference is factually correct and contextually appropriate.
Curriculum Relevance	0	The cultural element does not align with learning objectives or curriculum standards
	1	The cultural element partially aligns, but the connection is weak or tangential
	2	The cultural element strongly aligns with and enhances learning objectives
Hallucination	0	No fabricated or invented cultural elements
	1	Minor embellishments or assumptions about cultural practices
	2	Significant fabrication of cultural information not grounded in reality

Note: Higher scores indicate better performance.

```
Python
def generate_full_prompt(parameters):
    """
Generate a comprehensive culturally responsive prompt based on parameters"
    """
    country = parameters["country"]
    region = parameters["region"]
    grade = parameters["grade"]
    school_level = parameters["school_level"]
    subject = parameters["subject"]
    topic = parameters["topic"]
    # Construct a theoretically grounded prompt with all cultural dimensions
    prompt = f"""
    Develop a detailed, culturally responsive lesson plan for teaching {topic} within {subject}
    to {grade} students in {school_level} in {region}, {country}.
    === EDUCATIONAL CONTEXT ===
    - Class size: {parameters["class_size"]}
    - Lesson duration: {parameters["lesson_duration"]} minutes
    - Learning objectives: {parameters["learning_objectives"]}
```

Box 3. Prompt Generator Function

```
===CULTURAL RESPONSIVENESS FRAMEWORK
===For this lesson plan, you must thoroughly integrate the cultural context of {region}, {country}.
Access your knowledge about this region:
- Dominant language and linguistic features
- Cultural practices, traditions, and celebrations
- Indigenous knowledge systems
- Local artifacts, materials, and resources
- Community activities and daily life
- Belief systems and worldviews
Your lesson plan must incorporate all five dimensions of cultural responsiveness:
1. INDIGENOUS LANGUAGE INTEGRATION
- Include key terminology in both English and the dominant local language
- Provide pronunciations and translations for important concepts
- Incorporate language in authentic ways throughout the lesson
2. INDIGENOUS KNOWLEDGE SYSTEMS
- Connect scientific concepts to traditional knowledge in {region}
- Acknowledge and respect traditional ways of understanding
- Highlight the value of Indigenous approaches alongside academic content
3. CULTURAL CONTEXTUALIZATION
- Use examples relevant to students' lives in {region}, {country}
- Reference local practices, environments, and experiences
- Make abstract concepts concrete through cultural connections

4. CULTURALLY ALIGNED PEDAGOGY
- Incorporate teaching methods that align with cultural learning approaches
- Consider communication styles and classroom dynamics appropriate to the culture
- Design activities that honor cultural values and practices

5. CULTURALLY RESPONSIVE ASSESSMENT
- Create assessment approaches that respect cultural ways of demonstrating knowledge
- Provide multiple means of expression (oral, written, demonstrative)
- Connect assessment to community applications
=== LESSON PLAN STRUCTURE ===
Structure the lesson plan with the following sections:
1. Title and Overview (incorporating cultural context)
2. Learning Objectives (content, language, and cultural objectives)
3. Materials Needed (including locally available materials)
4. Prior Knowledge Required (including cultural knowledge)
5. Introduction/Hook (incorporating cultural elements)
6. Main Activities (step-by-step, with timing)
7. Assessment Methods (culturally responsive approaches)
8. Differentiation Strategies (addressing diverse learners)
9. Extension Activities (connecting to community and home)
10. Reflection Questions (for students and teachers)
Cultural Enrichment
"""
# Add any optional user-provided cultural information
    prompt += f"\n\nNote: The dominant local language is {parameters['dominant_language']}. Please incorporate this language throughout the lesson plan."
If "cultural_notes" in parameters:
    prompt += f"\n\nAdditional cultural context: {parameters['cultural_notes']}"

return prompt
```

Box 4. Cultural Responsiveness Framework

This process also ensures that educators have agency by recognizing that, while the system leverages GPT-4o's knowledge base, educators remain the cultural authorities for their specific contexts. The prompt allows educators to provide any cultural information or knowledge they might want to include in their lessons. This collaborative approach honors a human–AI partnership that produces more authentic educational content while preserving educators' decision-making authority in the culturally responsive lesson planning process.

The final function saves the generated lesson plans in both plain text (.txt) format for universal accessibility and Microsoft Word (.docx) format for professional presentation and editing. This dual-format approach ensures that educators can download and modify the materials as needed.

The complete implementation workflow integrated all components into a seamless prompt for the generation of culturally responsive lesson plans (see the Complete CRLP Script).

3.2.1.4. Baseline test of the script. To ensure that the computational environment was properly configured for AI-assisted lesson plan generation, a test script was executed within a Jupyter Notebook. The primary aim of this script was to validate the installation and interoperability of key Python libraries, including *OpenAI* for API access, the prompt, P*ython-docx* for Microsoft Word document creation, and Jupyter's file management for seamless output delivery. The test script prompted the GPT-4o model to generate a culturally responsive science lesson plan tailored for Grade 7 students in the Ashanti Region of Ghana, incorporating elements such as local language (Twi), indigenous knowledge, and contextually relevant activities.

The resulting output was successfully saved in both plain text (.txt) and Microsoft Word (.docx) formats, with direct download links provided within the notebook interface (see Appendix A for the test script and the downloaded lesson). The successful execution of this test confirmed the reliability of the prompt and the environment for performing the full script. The Word document generation component is particularly significant, as it transforms the raw lesson plan text into a professionally formatted document that teachers can immediately use in their practice. This addresses a critical usability concern, making the culturally responsive content accessible in a familiar format that aligns with educators' existing workflows.

3.2.2. Transition to custom GPT implementation

While our Python implementation demonstrated the effectiveness of theoretically grounded prompts in generating culturally responsive lesson plans, we recognized that many educators lack the technical expertise to use a Python-based system. This led us to translate our approach into a custom GPT implementation using OpenAI's ChatGPT platform, maintaining the theoretical integrity of our approach while eliminating technical barriers. The ISA prompt architecture implemented in the custom GPT (the CRLP) guided users through the same cultural parameter collection process but in a more accessible conversational interface.

3.2.2.1. Configuration of the Culturally Responsive Lesson Planner. The CRLP implemented the theoretical framework using the ISA prompt architecture. When users interact with the CRLP, they are welcomed with a conversational introduction that establishes rapport and explains the GPT's purpose in designing culturally responsive lessons. The system methodically collects essential cultural parameters, including country, region, student demographics (languages spoken, ethnicity), and community contexts (Nyaaba et al., 2024). Importantly, the prompt flow gives educators agency by asking whether they prefer to integrate cultural elements throughout the entire lesson or focus on specific sections, allowing for customized implementation of Gay's (2018) comprehensive integration principle.

To enhance functionality, we enabled web browsing, image generation, data analysis, and canvas creation capabilities, allowing the CRLP to access region-specific content and create more culturally appropriate educational materials (see Fig. 3). This deliberate configuration transforms the GPT from a generic lesson planner into a culturally responsive educational assistant that preserves educator decision-making authority while systematically implementing theoretical principles of culturally responsive pedagogy [The CRLP GPT is accessible at this link: *https://chatgpt.com/g/g-ei2C8xiFM-culturally-responsive-lesson-planner-crlp*]

3.2.2.2. Interactive semi-automated prompt implementation. Building on our prior use of an *Interactive Guided Dynamic* (IGD) prompt (Nyaaba et al., 2024), this study operationalizes an ISA prompt to balance theory-driven structure with teacher-guided customization. Whereas the IGD supported open-ended exploration through free-form dialogue, the ISA constrains that dialogue to a predefined, CRP-aligned schema. This design choice ensures replicability, evaluation fidelity, and consistent inclusion of CRP dimensions while preserving interactivity for local adaptation.

Architecture and user experience. Fig. 4 illustrates the ISA prompt architecture embedded within the CRLP. The interface opens with a concise, practitioner-friendly preamble that (a) states the tool's purpose, to co-construct culturally responsive lesson plans, and (b) previews the information requested. The system then collects contextual data through a short, sequenced interview consisting of targeted questions mapped to CRP constructs. Each question corresponds to a named slot in the underlying template, enabling semi-automation once values are provided.

Using sequenced data collection, the conversation proceeds in fixed stages:

1. Locale and setting: country → region → community/municipality (e.g., *Ghana → Ashanti Region → Kumasi*).
2. Community assets and knowledge: locally meaningful practices, proverbs, crafts, occupations, landforms, historical figures, and funds of knowledge.
3. Linguistic context: home and school languages (e.g., Akan/Twi), bilingual goals, translanguaging preferences.
4. Learner profile: grade/age band, prior knowledge, access constraints (e.g., devices, connectivity, cost).
5. Curricular anchors: subject strand/topic, standards or learning targets, assessment mode (formative/summative).
6. Representation and relevance: examples, scenarios, and named communities to be centered; sensitivities to avoid.

At each stage, the CRLP acknowledges and reflects the provided context (e.g., recognizing the cultural heritage of the Ashanti Region in Ghana and indicating that Akan culture and Twi will be incorporated where appropriate). This brief reflective confirmation both validates teacher input and allows quick correction before the pipeline advances.

Once all necessary information is collected, the ISA prompt engine enables the CRLP to assemble theory-based lesson plans with the following sections: (a) goals/standards, (b) culturally anchored hooks and examples, (c) learning activities aligned to community practices and funds of knowledge, (d) language supports (terminology in English and the local language; prompts for translanguaging), (e) differentiation and no/low-cost materials, and (f) assessment tasks with CRP-aligned success criteria. The semi-automation ensures that core CRP elements cannot be omitted; the interactive steps ensure that those elements are localized rather than generic. This conversational flow exemplifies how the theoretical frameworks were translated into an accessible, step-by-step process that collects the necessary cultural parameters while maintaining a user-friendly experience for educators without technical expertise in prompt engineering (see Appendix B).

3.2.2.3. Output consistency. Through an iterative refinement process, we rigorously tested the CRLP's lesson plan generation capabilities, conducting at least three complete test cycles. During these tests, we analyzed the consistency and quality of outputs, finding that the CRLP produced remarkably similar lesson plans across multiple generations with nearly identical cultural elements, indigenous language terms, and pedagogical approaches. While the three generated lesson plans showed high consistency in their cultural responsiveness, we selected the most comprehensive version (copied and pasted) for our comparative analysis, which ensured that we were evaluating the CRLP's optimal performance against the standard prompt. This testing process confirmed both the reliability of our theoretically grounded prompt architecture and the stability of the CRLP's cultural integration capabilities.

3.3. Selection and comparison of the standard prompt with the interactive semi-automated prompt

The OpenAI-recommended lesson plan prompt was used in our comparative evaluation. Therefore, we selected a standard prompt from six suggested prompt options provided in the OpenAI playground (see Appendix C) to generate culturally responsive lesson plans with the base GPT-4o, for comparison with those generated by the CRLP. The playground served as an experimental interface for testing and refining prompts for OpenAI's language models, offering multiple template options for various use cases. After systematically testing all six suggested lesson plan prompts, we selected the top-performing option (the first prompt) as our comparison baseline, as it generated lesson plans with the most culturally relevant elements that met our study's objective (see Appendix D for the Standard Prompt). Unlike the other five suggested prompts generated in the Playground, this prompt specifically asked the GPT model not just to acknowledge but to explicitly integrate students' home languages, community traditions, and lived experiences. Additionally, the requirement to identify at least one culturally relevant example and corresponding activity ensured a concrete connection to students' cultural backgrounds, thus grounding theoretical principles of culturally responsive education in practical classroom strategies. This selected prompt represents OpenAI's best suggested approach for generating culturally responsive lesson plans.

The comparison of the two prompt systems was situated within four criteria (see Table 1). Even though both involve interactions with users, the ISA interaction is more dynamic and iterative compared to the Standard OpenAI prompts, which remain static and one-time. Again, when it comes to cultural context customization, the ISA offers high and detailed cultural customization that is guided and structured, compared to the standard system, which appears moderate and depends entirely on what users can articulate. With user engagement and validation, the ISA is extensive (with multiple touchpoints for feedback), while the standard is minimal (with no validation steps). Finally, for complexity and usability, the ISA is more complex and involves more time due to its multiple interactions, whereas standard prompts are straightforward and save time.

These four criteria highlight the fundamental trade-offs between the two systems: ISA prioritizes depth and accuracy through its iterative

Table 4
Evaluation of the culturally responsive lesson plans.

Metric	Reviewer 1		Reviewer 2		Average Ratings	
	Standard	CRLP	Standard	CRLP	Standard	CRLP
Total Cultural Elements	23	49	19	48	21	48.5
Mean Accuracy Rating (0–2 / %)	1.2 (60%)	1.8 (90%)	1.3 (65%)	1.8 (90%)	1.2 (60%)	1.8 (90%)
Mean Relevance Rating (0–2 / %)	1.8 (90%)	2.0 (100%)	1.3 (65%)	2.0 (100%)	1.3 (65%)	2.0 (100%)
Cultural Hallucinations	1.0 (50%)	1.0 (50%)	0.0 (0%)	0.5 (25%)	0.5 (25%)	0.75 (37.5%)

Note: Ratings were based on a 0–2 scale (0 = not present / 0%; 1 = partially present / 50%; 2 = fully present / 100%). Higher scores indicate better performance.

approach, while standard prompts prioritize efficiency and simplicity for users who need quick responses.

3.3.1. The generated lesson plans

We generated two lesson plans for teaching the states of matter to 7th-grade students in the Ashanti Region of Ghana, where Asante Twi is the dominant local language. One of the lsson plans was created using the standard GPT-4o model (see the direct copy-and-paste version in Appendix E), and the other was created using our CRLP (see the direct copy-and-paste version in Appendix F). Both tools were given the same topic and grade level, ensuring a fair comparison. The lesson plans were then evaluated by two Ghanaian educators with expertise in culturally responsive pedagogy inthe educational context of the Ashanti Region in Ghana.

3.3.1.1. Analysis. To address the first research question, we engaged the two expert reviewers with deep familiarity in Ghanaian education and culturally responsive pedagogy to evaluate the two lesson plans. In line with accepted practices in qualitative evaluation, our reviewers independently rated each lesson plan using the rubric. This double-coding was conducted without consensus meetings, as the emphasis was on transparency, analytical depth, and justification rather than inter-rater agreement (O'Connor & Joffe, 2020).

Their evaluation was guided by a rubric derived from the study's theoretical framework, covering three dimensions: *Accuracy, Curriculum Relevance, and Hallucination* (see Table 3). The *Accuracy dimension* assessed whether the cultural reference was factually correct and reflective of authentic Ghanaian practices, for example, referencing a *bukyea* (traditional three-stone stove) in lessons on heat transfer. *Curriculum Relevance* examined whether the cultural reference meaningfully supported the intended science learning objectives. *Hallucination* refers to any instance where fabricated or culturally misplaced information is present in the GenAI-generated content.

The reviewers began by identifying distinct cultural elements embedded in each lesson plan. Each reviewer independently scored the two lesson plans using the rubric on a 0–2 scale across the three dimensions. The total number of cultural elements identified per lesson plan was calculated, and average scores for each dimension were computed across the two reviewers. To enhance interpretability, each categorical rating on the 0–2 scale was converted into a corresponding percentage value (0 = 0%, 1 = 50%, 2 = 100%), providing a clearer representation of the extent to which each criterion, accuracy, relevance, and the cultural appropriateness of the identified elements, was fully met.

In addition to numerical ratings, reviewers' comments and supporting evidence from the lesson plans were analyzed thematically. We adapted Clarke and Braun's (2016) thematic analysis approach, though not comprehensively, as this was not a full qualitative research study. Instead, we employed a streamlined deductive approach using our theoretical framework as an analytical guide. Our analysis followed a focused coding procedure to identify relevant keywords and phrases from reviewer comments and categorize them within our predetermined theoretical framework dimensions. This deductive approach allowed the theoretical framework to serve as the basis for coding and theme identification. Rather than conducting all six phases of thematic analysis, we familiarized ourselves with the review comments through initial reading, identified key words and phrases relevant to cultural elements, sorted these under our framework categories, and noted outlier comments that revealed unexpected findings.

4. Results

Table 4 presents expert reviewer evaluations comparing both tools. The CRLP significantly outperformed the standard GPT-4o across all metrics. Most notably, CRLP-generated plans contained more than double the cultural elements (48.5 vs. 21), indicating the standard prompt's limited capacity to incorporate cultural content without structured theoretical guidance. The standard prompt's lower accuracy (1.2 out of 2.0 [60%] vs. 1.8 out of 2.0 [90%]) and relevance (1.3 out of 2.0 [65%] vs. 2.0 out of 2.0 [100%]) ratings likely stem from its cultural omissions. Both approaches showed similar levels of cultural hallucinations (0.5 out of 2.0 [25%] for the Standard prompt and 0.75 out of 2.0 [37.5%] for the CRLP-GPT), suggesting that the standard prompt's primary limitation is cultural omission rather than fabrication of cultural content.

Based on the evaluation presented in Table 4, our analysis revealed significant differences in cultural responsiveness across multiple dimensions. The following sections present the detailed findings from the qualitative comments.

4.1. Indigenous language integration

Indigenous language integration revealed one of the most pronounced differences between the standard prompt within the base GPT-4o and the CRLP. The CRLP incorporated approximately three times more indigenous language elements than the standard prompt output. More significantly, the CRLP integrated indigenous language throughout the instructional process rather than isolating it in a dedicated vocabulary section.

Reviewer 1 noted:

> *The standard prompt lesson plan contains basic Twi vocabulary but uses it primarily in isolation rather than integrating it throughout instruction. The ISA approach demonstrates more natural language integration, with Twi being used meaningfully throughout all phases of the lesson, including greetings, instructions, explanations, and assessment.*

Highlighted evidence from the standard prompt lesson plan:

"Key Vocabulary (English ↔ Twi) English Term Twi Equivalent Matter Ahoɔdenfɛfɛw Solid Den adeɛ Liquid Nsuo te sɛ Gas Mframa Change of State Nsesa ɛfiri baako kɔ baako." (Reviewer 1)

Reviewer 1 further noted that, by contrast, the CRLP integrated indigenous language naturally throughout the lesson structure, as seen in this introduction sequence:

"Greet learners in Asante Twi: *Mema mo akye! Yɛbɛkasa afa adeɛ a yɛfrɛ no States of Matter ho. Ask: Ɛhe na moahu nsuo anaa ngo da ho? Na dɛn na ɛba mu sɛ wode nsuo bɛde gu fridge mu?"*

Regarding accuracy, Reviewer 2 identified potential inaccuracies in the standard approach: "The term *Ahoɔdenfɛfɛw* for 'matter' in the standard prompt lesson plan appears unusual and may have been a mistranslation." The CRLP demonstrated higher accuracy but still

contained occasional spelling variations.

4.2. Indigenous knowledge systems

Both the standard prompt lesson plan with the base GPT-4o and the CRLP lesson plan showed evidene of indigenous knowledge, but with notable differences in specificity and relevance. The CRLP lesson plan had more elements of indigenous knowledge specifically aligned with the Ashanti regional context.

Reviewer 1 observed:

> *Both approaches demonstrate the incorporation of indigenous knowledge systems, but with notable differences. The standard approach references* fufu *pounding and* pito *brewing, though* pito *is more commonly associated with Northern Ghana than the Ashanti region. The CRLP uses culturally appropriate metaphors and examples that are specific to the Ashanti context and more immediately accessible to students. The standard prompt included valuable traditional examples but with some regional incongruities.*

Highlighted evidence from the standard prompt lesson plan:

"Pito brewing involves boiling, fermentation, and evaporation, clearly showing liquid-to-gas transformations." (Reviewer 1)

Highlighted evidence from the CRLP lesson plan:

"Explain using metaphor: *Sɛ yɛfa atadeɛ ahyehyɛ ho* (solid), *nsuo a ɛda basin mu* (liquid), and *cooking gas a ɛwɔ cylinder mu* (gas)." (Reviewer 1)

4.3. Cultural contextualization

The CRLP lesson plan demonstrated stronger cultural contextualization, with more examples (8 vs. 6) that were specifically relevant to the Ashanti regional context. These contextual elements were more integrated throughout the lesson structure rather than appearing as standalone cultural references.

Reviewer 2 explained:

> *Cultural contextualization in the CRLP lesson plan is more thoroughly integrated into the entire lesson structure. The examples provided are highly relevant to students' lived experiences in the Ashanti region. The Standard approach makes good attempts at contextualization but includes some examples that may be less familiar to students in this region.*

Highlighted evidence from the standard prompt lesson plan:

"*Fufu pounding and pito* brewing are used to demonstrate changes in the state of matter. Fufu pounding: cassava and plantain are solids that become semi-solid through pounding and heat. *Pito* brewing: involves boiling, fermentation, and evaporation, clearly showing liquid-to-gas transformations."

Highlighted evidence from the CRLP lesson plan:

Local examples in Asante Twi:

"Solid (*Dadeɛ/Abo*): *ɛpono, dadeɛ nkuto, aboɔ, ntadeɛ*
Liquid *(Nsuo/Nsa/Ngo): nsuo, ngo, nsa, akpeteshie*
Gas (*Mframa/Ahotɔ*): *mframa*, cooking gas (LPG), *nkuto mframa*." (Reviewer 2)

4.4. Culturally aligned pedagogy

Both the standard prompt within the base GPT-4o and the CRLP demonstrated awareness of culturally appropriate pedagogical approaches, but the CRLP lesson plan showed better alignment with typical Ghanaian classroom practices and resource constraints.

Reviewer 1 commented:

> *Both lesson plans incorporate culturally appropriate pedagogical approaches, but the CRLP approach shows greater alignment with typical Ghanaian classroom practices. The Standard approach suggests valuable activities like community exhibitions and elder visits, but these may be logistically challenging in many Ghanaian schools. The ISA approach offers more immediately implementable strategies that work within typical resource constraints.*

Highlighted evidence from the standard prompt lesson plan:

"Create a community exhibition titled *Science in Our Traditions* where students display their drawings and narrate the science behind common practices in Asanteman." (Reviewer 1)

Highlighted evidence from the CRLP lesson plan:

"Group 1: Hold and describe ice cubes (*ɛyɛ den, ɛwɔ shape – Solid*).
Group 2: Pour and observe palm oil (*ɛtumi flow, ɛyɛ Liquid*).
Group 3: Release air from the balloon (*ɛkɔ soronko, ɛyɛ Gas*).

Let each group report in Asante Twi what they observed." (Reviewer 1)

4.5. Culturally responsive assessment

The CRLP lesson plan showed a robust culturally responsive assessment design, with assessment strategies specifically developed to incorporate cultural and linguistic elements. The standard prompt mentioned cultural assessment but with less specific integration.

Reviewer 2 noted:

> *Assessment strategies in the CRLP lesson plan demonstrate a better understanding of culturally responsive evaluation. The bilingual questioning, cultural matching exercises, and home connection assignments provide authentic ways to assess understanding while honoring students' cultural context. The Standard prompt identifies assessment strategies, but with less cultural specificity.*

Highlighted evidence from the standard prompt lesson plan:

"Assessment • Observation of group activities • Completion and quality of worksheet • Oral participation and use of Twi terms."

Highlighted evidence from the CRLP lesson plan:

"Assessment/Evaluation (10 min):

1. Oral Questions (in Twi/English):
   - *Yɛfrɛ dɛn adeɛ a ɛtumi flow te sɛ nsuo?*
   - *Dɛn na ɛbɔ mframa ho din wɔ brofo mu?*
2. Classwork: Match local items to their states:
   - *Nsa*, *ɛpono*, cooking gas → Liquid, Solid, Gas
3. Take-Home Assignment: Draw three household items and label their state of matter in English and Twi." (Reviewer 2)

4.6. Contextual resources

A significant distinction between the standard prompt within GPT-4o and the CRLP is the inclusion of explicit, contextualized resources. The CRLP included contextualized and suggested resources that were entirely absent from the standard prompt output. Even though the standard prompt included general materials needed for the lesson, it lacked any explicit suggested resources section to support teacher implementation.

Reviewer 2 emphasized:

> *The CRLP includes a section of suggested resources specifically tailored to the lesson context, including references to "CK−12 Interactive Science Concepts," "YouTube video explaining States of Matter for African learners," and "Science Made Easy for Ghanaian JHS—Book." These contextualized resource suggestions provide immediate practical support for teachers implementing the lesson, a feature completely absent from the standard approach.*

4.7. Teacher agency and collaboration

The CRLP explicitly invited teacher input and modification, a feature completely absent from the standard prompt approach. The standard prompt provided no such explicit invitation for teacher feedback or modification, presenting the lesson plan as a finished product rather than a collaborative process. This fundamental difference reflects how the ISA approach intentionally preserves teacher decision-making authority within the CRLP process.

Reviewer 1 noted:

> *The CRLP concludes by asking, "Would you like to review or modify any part of the lesson before I wrap it up in a printable format?" This invitation for educator input positions teachers as active collaborators rather than passive recipients, acknowledging their expertise and authority in the lesson development process.*

4.8. Developer accessibility

A unique strength of custom GPTs compared to base models emerged in the CRLP's explicit developer contact information.

Reviewer 1 highlighted:

> *"You can contact Anonymized [creator] for access: Email: Anonymized, LinkedIn: Anonymized."*

This feature, entirely absent from the base GPT, establishes a direct feedback channel between users and developers. By embedding creator contact information directly in the output, the CRLP creates a transparent development relationship that GPT models typically lack, facilitating ongoing improvement based on educator experiences. This accessibility transforms the tool from a static product into an evolving system responsive to real-world implementation feedback, further reinforcing the collaborative philosophy underlying the CRLP lesson plan while addressing a fundamental limitation of generic GPT implementations.

5. Discussion

The findings of this study support the view that GenAI is capable of generating lesson plans that incorporate structured sequences aligned with established pedagogical theories. This is consistent with Baytak's (2024) study, which found that GenAI-generated lesson plans reflected concepts grounded in diverse learning theories and models, highlighting GenAI tool's adaptability to different pedagogical orientations. The findings further support Nyaaba et al.'s (2024) study, confirming that custom GPTs with theoretical frameworks can effectively enhance culturally responsive pedagogy. Nyaaba et al. (2024) demonstrated that their CRSciA-generator was able to develop culturally responsive assessments for use in diverse national contexts by incorporating culturally relevant elements such as indigenous knowledge and languages. Similarly, our CRLP demonstrated significant improvements in cultural elements (48.5 vs. 21), accuracy (1.8 vs. 1.2), and relevance (2.0 vs. 1.3) compared to standard prompts, with a particularly strong enhancement in indigenous language integration.

The findings also support Lee and Zhai's (2024) study regarding accuracy concerns, aligning with their observation that some lesson plan elements werehallucinated when preservice teachers use ChatGPT to generate them. In our study, this was evidenced by Reviewer 2's identification of potentially inaccurate indigenous language terms such as *Ahoɔdenfɛfɛw* for "matter," which supports Lee and Zhai's (2024) findings about accuracy limitations. However, despite these occasional inaccuracies, the findings indicate that the CRLP-GPT enhanced cultural representation but also exhibited a slight increase in cultural hallucinations (from 0.5 to 0.75) on the rating scale, highlighting that theoretically grounded prompting strategies can improve contextual depth without fully eliminating generative distortion (Lee and Zhai's, 2024).

Moreover, the findings align with Peikos and Stavrou's (2025) exploratory study, which found that incorporating PCK elements into prompts, using layered prompting strategies, and providing reference texts to ChatGPT can enhance the quality of AI-generated lesson plans. However, their approach was often cumbersome for educators (Peikos & Stavrou, 2025). In contrast, our ISA prompt within the CRLP offers a more streamlined solution, allowing educators to benefit from theoretically informed strategies without needing to craft prompts or provide extensive reference materials. This addresses a practical implementation barrier for educators with limited time or prompt engineering skills (Nyaaba et al., 2024).

Another intriguing alignment appears in Peikos and Stavrou's (2025) emphasis on teacher–ChatGPT co-design of lesson plans, which directly supports our finding on teacher agency. As Reviewer 1 observed, the CRLP concludes by asking: *"Would you like to review or modify any part of the lesson before I wrap it up in a printable format?"* This invitation for modification positions teachers as active collaborators rather than passive recipients of GenAI output. The co-design approach advocated by Peikos and Stavrou (2025) and the concerns raised by Bloom and Francis (2024) both underscore the need to address bias in AI integration. Our findings demonstrate that the CRLP, when used with the ISA prompt architecture, equips teachers with the agency to recognize and mitigate such biases, thereby preserving their decision-making authority while fostering more equitable learning opportunities.

The capability of the CRLP also addresses Kitsara's (2022) concerns about unequal access to AI resources and skills across different geographies, offering a more inclusive approach to AI implementation in education. The customized approach makes culturally responsive AI more accessible to diverse contexts. Although we did not include curriculum materials in the configuration parameters, the customization features allow the CRLP to draw knowledge from uploaded materials, which could further address digital divides if we had incorporated locally relevant resources (OpenAI, 2024a; OpenAI, 2024b).

However, important limitations regarding generalizability and structural challenges must be acknowledged. While our approach offers generalizability potential, as the prompts guide users in leveraging GPT's knowledge base to draw specific outputs relating to their locality, we acknowledge Nyaaba et al.'s (2024) concern that users might need to include more detailed contextual information when necessary. Even with the CRLP's improvements, cultural hallucinations persisted at an average of 0.75 out of 2.0 (37.5%), indicating that uneven training data across cultural contexts continue to limit GenAI's contextual accuracy (Watson et al., 2024). This situation is only partially addressed through the co-design agency of the CRLP, which allows educators to include additional cultural elements to improve their lesson plans. Nonetheless, the structural limitations in AI training data remain a significant challenge that incremental improvements in prompting cannot fully overcome (Yogarajan et al., 2024).

6. Conclusion

In this study, we demonstrated that generic GenAI lesson plans often fail to capture the cultural and contextual realities of classrooms, particularly in underserved communities (e.g., African contexts). Adopting a design science research approach and grounding our development in CRP, we designed and evaluated a CRLP GPT. The CRLP, built upon a semi-interactive prompt schema, systematically collects cultural and educational parameters before generating lesson plans. This approach ensured that cultural responsiveness was embedded across objectives, activities, and assessments rather than treated as an afterthought. Our comparative evaluation of Grade 7 science lesson plans in Ghana's Ashanti Region confirmed that the CRLP integrates local language, indigenous knowledge systems, and culturally aligned pedagogy more effectively than standard GPT prompts. Specifically, expert reviewers found that lesson plans generated with the CRLP incorporated more than double the cultural elements of standard GPT-4o outputs,

with higher ratings for accuracy and curriculum relevance. Examples such as the integration of Asante Twi vocabulary and culturally relevant practices like fufu preparation or metaphors rooted in everyday community life illustrate how the CRLP reflects students' lived realities. Beyond improving the cultural responsiveness of lesson content, the CRLP's co-development process represents a Human-in-the-loop system that, through interactive prompts and engagement between the teacher and the CRLP, has the potntial to foster teacher agency and GenAI literacy by directly involving educators in the prompt process. However, cultural hallucinations were slightly higher in the CRLP-GPT compared with the standard GPT-4o prompt. These findings underline that custom GPTs grounded in educational theories such as CRP can generate materials that are both contextually relevant and pedagogically sound, while holding significant potential to strengthen teachers' knowledge and practice.

7. Limitations and future directions

There are several limitations in this study that must be addressed in future research. Although the co-design agency feature allows educators to embed cultural elements, this does not entirely mitigate the risk of hallucinations inherent in GenAI outputs, particularly due to limited training data representing many underrepresented cultural contexts. The structural biases in LLMs remain a significant challenge, which underlines the need for continuous dataset diversification and iterative user feedback loops. Furthermore, because the Ashanti Region and Asante Twi were the focus of this study, the findings cannot be generalized to all Ghanaian languages or regions, particularly as Asante Twi is among the most widely spoken languages in Ghana. With Ghana's 16 regions and more than 80 indigenous languages, broader testing across different regions and languages will be needed to evaluate the CRLP's adaptability to the Global South's full linguistic and cultural diversity. Looking ahead, this study points to the need for broader validation and classroom-based testing of culturally responsive GenAI-generated lesson plans. Future research should compare CRLP outputs with those of human generated lesson plans and examine their impact on student learning outcomes across diverse contexts.

CRediT authorship contribution statement

Matthew Nyaaba: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Xiaoming Zhai:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

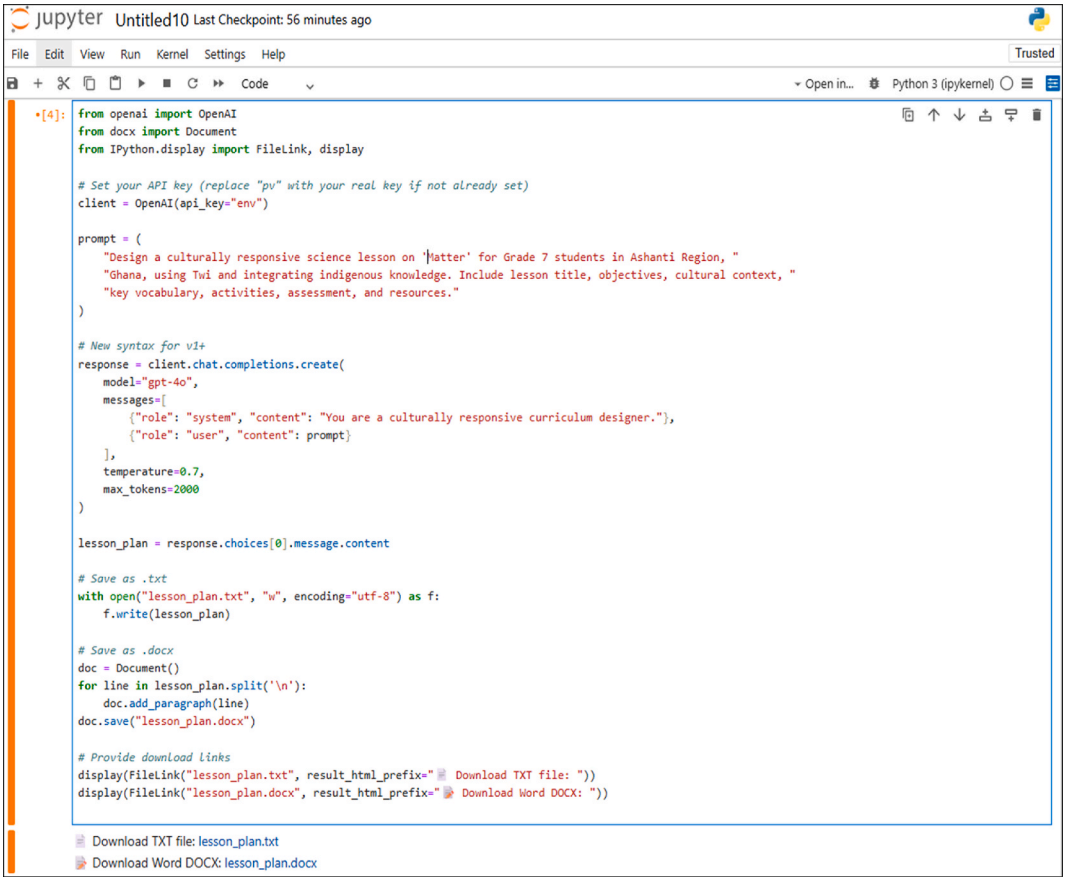
Acknowledgements

This material is partially based upon work supported by the National Science Foundation under Grant No. 2101104. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

Appendix A

Link to the Theory-informed Python Script for the CRLP: https://github.com/matthewnyaaba/CRLP-Python-Script

The Executed Test Script of Prompt


jupyter Untitled10 Last Checkpoint: 56 minutes ago
File Edit View Run Kernel Settings Help
Trusted
Code
Open in...
Python 3 (ipykernel)
•[4]:
from openai import OpenAI
from docx import Document
from IPython.display import FileLink, display
Set your API key (replace "pv" with your real key if not already set)
client = OpenAI(api_key="env")
prompt = (
"Design a culturally responsive science lesson on 'Matter' for Grade 7 students in Ashanti Region, "
"Ghana, using Twi and integrating indigenous knowledge. Include lesson title, objectives, cultural context, "
"key vocabulary, activities, assessment, and resources."
)
New syntax for v1+
response = client.chat.completions.create(
model="gpt-4o",
messages=[
{"role": "system", "content": "You are a culturally responsive curriculum designer."},
{"role": "user", "content": prompt}
],
temperature=0.7,
max_tokens=2000
)
lesson_plan = response.choices[0].message.content
Save as .txt
with open("lesson_plan.txt", "w", encoding="utf-8") as f:
f.write(lesson_plan)
Save as .docx
doc = Document()
for line in lesson_plan.split('\n'):
doc.add_paragraph(line)
doc.save("lesson_plan.docx")
Provide download links
display(FileLink("lesson_plan.txt", result_html_prefix=" Download TXT file: "))
display(FileLink("lesson_plan.docx", result_html_prefix=" Download Word DOCX: "))
Download TXT file: lesson_plan.txt
Download Word DOCX: lesson_plan.docx

Test Script Output From the Python + OpenAI API

Lesson Title: Understanding Matter Through Indigenous Knowledge
Objectives:
1. Students will be able to define matter and identify its states using both scientific and local terms.
2. Students will explore the cultural significance of matter in the Ashanti Region, specifically how indigenous knowledge relates to understanding matter.
3. Students will demonstrate the ability to classify objects around them into different states of matter.
Cultural Context:
The Ashanti Region of Ghana is rich in history and culture, with a deep connection to indigenous knowledge and practices. The traditional Ashanti belief system involves a profound understanding of natural elements, which can be related to the scientific concept of matter. By integrating Twi language and indigenous knowledge, students will gain a deeper understanding of the science of matter in a culturally relevant context.
Key Vocabulary:
- Matter: Adeɛ
- Solid: Dɛm
- Liquid: Nsuo
- Gas: Mframa
- Indigenous Knowledge: Nkwanso Nyansa
- States of Matter: Adeɛ ahorow
**Activities:*
1. **Introduction to Matter (Class Discussion in Twi and English):**
  - Begin with a discussion about the concept of matter ("Adeɛ") and its importance in the world around us.
  - Use examples from everyday life, such as water ("Nsuo"), air ("Mframa"), and stones ("Dɛm").
  - Discuss how these examples are perceived in the local culture and their uses.
2. **Exploration of States of Matter (Group Activity):**
  - Divide students into groups and provide materials such as water, stones, and balloons.
  - Have each group observe and classify the materials into solid, liquid, and gas.
  - Encourage students to use Twi to describe their observations and share indigenous stories or proverbs related to these materials.
3. **Indigenous Knowledge and Science (Storytelling Session):**
  - Invite a local elder or storyteller to share tales or proverbs that relate to the concept of matter.
  - Discuss how these stories reflect the Ashanti understanding of natural elements.
  - Relate these stories back to scientific concepts to show the link between indigenous knowledge and modern science.

Continued:

4. **Hands-On Experiment (Simple Demonstration):**
   - Conduct a simple experiment to demonstrate the states of matter, such as melting ice to show the transition from solid to liquid.
   - Use this as an opportunity to discuss the local significance of water and how it is managed in the community.
Assessment:
- **Class Participation:** Observe students' participation in discussions and group activities, noting their ability to use key vocabulary in Twi and English.
- **Group Presentation:** Each group presents their findings from the exploration activity, using both scientific terms and indigenous knowledge.
- **Reflection Essay:** Ask students to write a short essay reflecting on what they learned about matter and how it relates to their culture.
Resources:
- Visual aids depicting states of matter
- Materials for experiments (water, stones, balloons)
- Local stories and proverbs (obtained from community elders or local libraries)
- Access to a speaker of Twi to assist in language integration
This lesson plan aims to bridge scientific understanding with cultural context, ensuring that students in the Ashanti Region can connect their learning to their heritage and everyday experiences.

Complete the Culturally Responsive Lesson Plan

Title and Overview
****Lesson Title:** Exploring Matter through the Lens of Ashanti Culture**
****Overview:** This lesson introduces students to the properties and states of matter, integrating Ashanti cultural knowledge and local practices to make learning relevant and engaging. Students will explore how traditional activities like pottery and weaving relate to scientific concepts of matter.**
Learning Objectives
1. Students will understand and describe the different states of matter: solid, liquid, and gas.
2. Students will identify and explain the properties of matter.
3. Students will relate indigenous knowledge from Ashanti culture to scientific concepts of matter.
Materials Needed
- Samples of Kente cloth (solid)
- Water (liquid)
- Local clay (solid)
- Charcoal from traditional stoves (solid)
- Incense for visualizing gas
- Adinkra symbols printed on cards
- Posters of cocoa farming and gold mining processes
Prior Knowledge Required
- Basic understanding of states of matter
- Familiarity with local cultural practices and artifacts
Introduction/Hook (5 minutes)
Begin the lesson by showing a piece of Kente cloth and asking students to describe its properties (solid, flexible, colorful). Connect this to our exploration of matter. Explain how the weaving process of Kente involves understanding the properties of the threads, which are made from solid materials.

Continued:

Main Activities

Activity 1: States of Matter and Local Examples (15 minutes)

- **Discussion: Describe the three states of matter (solid, liquid, gas). Use local examples: Kente cloth (solid), water from River Bosomtwe (liquid), and smoke from incense (gas).**

- **Demonstration: Burn incense to visualize gas and discuss its properties.**

- **Cultural Connection:** Talk about how incense is used in Adae ceremonies to honor ancestors, and link to the concept of gas.

Activity 2: Properties of Matter and Indigenous Knowledge (20 minutes)

- **Hands-On Activity:** Divide students into small groups. Provide each group with clay and water.

- **Task: Transform clay (solid) into a small pot, discussing its properties as they work.**

- **Discussion: Relate this to pottery and how the Ashanti people have traditionally understood the properties of clay to create strong, durable pots.**

- **Indigenous Knowledge Example:** Explain traditional medicine preparation, highlighting how understanding the properties of different plants (solid matter) is crucial in creating effective remedies.

Activity 3: Observing Changes in Matter (10 minutes)

- **Experiment: Heat a piece of cocoa butter until it melts, showing a change from solid to liquid.**

- **Discussion: Relate to cocoa farming, exploring how understanding the melting properties of cocoa is essential in producing cocoa products.**

- **Cultural Insight:** Discuss cocoa's role in the Ashanti economy and culture.

Assessment Methods

- **Formative Assessment:** Observe group discussions and participation during activities.

- **Summative Assessment:** Short quiz at the end of the lesson asking students to identify examples of solids, liquids, and gases from Ashanti culture and explain their properties.

Differentiation Strategies

- **Language Support:** Provide key vocabulary in both Twi and English.

- **Visual Aids: Use pictures and real objects, like Kente and clay, to support diverse learning styles.**

- **Peer Support: Encourage students to work in mixed-ability groups to support each other.**

Extension Activities

- Research and present on a local industry (e.g., gold mining) and its use of matter.
- Create a model demonstrating how traditional weavers understand the properties of threads.

Reflection Questions

1. How does understanding the properties of matter help in traditional Ashanti crafts and industries?

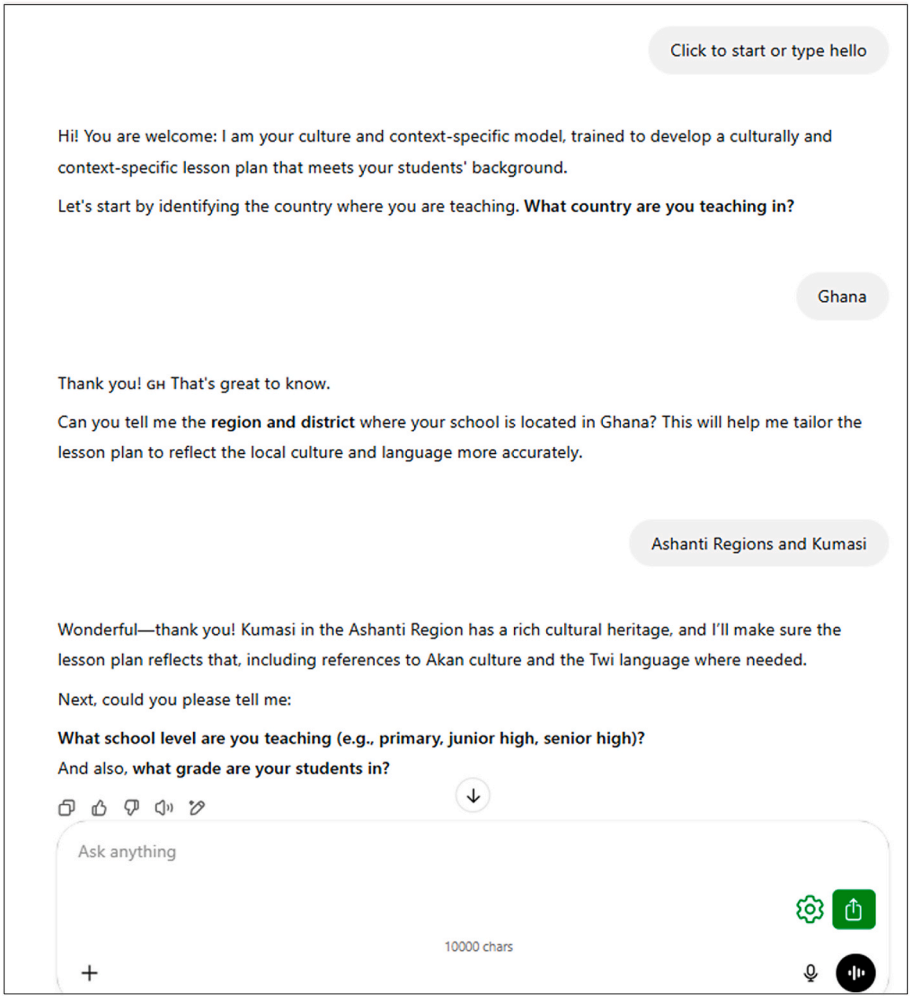
2. Can you think of another cultural practice that involves understanding different states of matter?

3. How do traditional beliefs and practices enhance our understanding of scientific concepts?

By integrating Ashanti cultural practices and local knowledge, this lesson plan aims to deepen students' understanding of scientific concepts while honoring their cultural heritage.

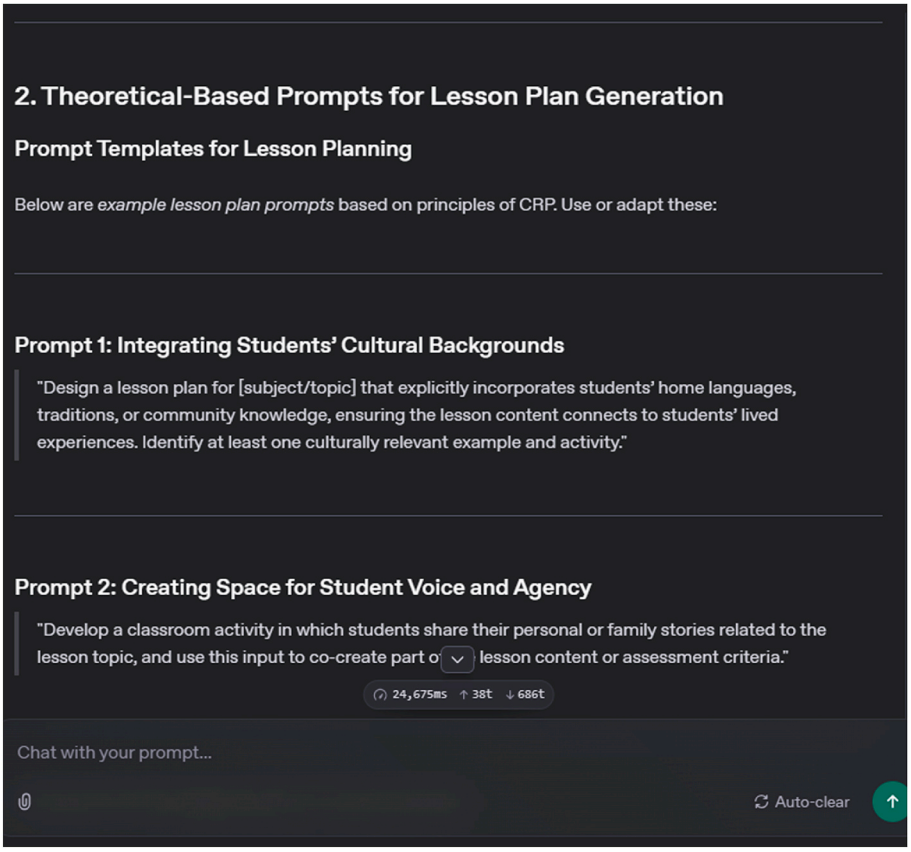
Appendix B

CRLP in Action



Appendix C

Standard Prompt Suggestions with the OpenAI Playground



Appendix D

OpenAI Standard Prompt Selected:

Design a lesson plan for [subject/topic] that explicitly incorporates students' home languages, traditions, or community knowledge, ensuring the lesson content connects to students' lived experiences. Identify at least one culturally relevant example and activity.

Appendix E

Standard Prompt Culturally Responsive Lesson Plan Output

Here is a Culturally Responsive Science Lesson Plan on "Matter" designed for Grade 7 students in Kumasi, Ashanti Region, Ghana. This lesson intentionally incorporates students' home language (Twi), local traditions, and community knowledge.

Subject: Integrated Science
Grade Level: Junior High School (JHS 1)
Topic: Introduction to Matter
Duration: 60 minutes
Language of Instruction: English with integration of Twi
Theme: Connecting Scientific Concepts with Indigenous Knowledge

---

Lesson Title: *"What Makes Up Our World: Exploring Matter through the Ashanti Lens"*

Lesson Objectives
By the end of the lesson, students will be able to:

1. Define matter as anything that has mass and occupies space.
2. Identify and describe the three main states of matter with local examples.
3. Use Twi terms to describe states of matter and commonly observed substances.
4. Analyze a cultural practice (e.g., pito brewing or fufu pounding) to explain the transformation of matter.
5. Relate the concept of matter to everyday life and community practices in Kumasi.

Key Vocabulary (English ↔ Twi)

English Term	Twi Equivalent
Matter	Ahoɔdenfɛfɛw
Solid	*Den adeɛ*
Liquid	*Nsuo te sɛ*
Gas	*Mframa*
Change of State	*Nsesa ɛfiri baako kɔ baako*

Materials Needed

- Calabash, pito brewing pot (or video of the process), mortar and pestle
- Water, ice, heated kettle
- Flashcards with Twi and English science terms
- Worksheet (includes drawing and reflection questions)

Culturally Relevant Example
Fufu pounding and pito brewing are used to demonstrate changes in the state of matter:

- *Fufu pounding*: Cassava and plantain are solids that become semi-solid through pounding and heat.
- *Pito brewing*: Involves boiling, fermentation, and evaporation—clearly showing liquid-to-gas transformations.

Lesson Procedure
Introduction (10 minutes)

- Greet students in Twi and engage them with a riddle:
  "Me yɛ den, me ntumi nsrɛ, na me ho yɛ hare. Me yɛ he?" (Answer: Solid)
- Ask students: "What do you know about what things are made of?" Use local examples like *nkuto* (shea butter), water from the borehole, or *aburo* (maize).

Direct Instruction (15 minutes)

- Define matter and its states using real-world, culturally familiar objects.
- Use Twi and English flashcards for reinforcement.
- Introduce transitions between states using a kettle (water to steam) and an ice melting demonstration.

Guided Practice (15 minutes)
Group Activity:

- In small groups, students act out matter transitions using local contexts.
- Example:
  Group 1 – *Fufu Pounding*
  Group 2 – *Pito Brewing*
  Group 3 – *Water Cycle using Rainwater Harvesting*

Each group presents their demonstration and identifies solids, liquids, gases, and changes of state, incorporating Twi vocabulary.

Independent Practice (10 minutes)

- Students complete a worksheet where they:
  - Match English-Twi science terms
  - Identify matter states in their home environment
  - Draw a traditional practice that involves matter transformation (e.g., palm oil processing)

Closure (5 minutes)

- Review vocabulary aloud in Twi and English.
- Ask:
  "Where in your home do you see these changes happening?"
  "How do our local practices show science in action?"

Assessment

- Observation of group activities
- Completion and quality of the worksheet
- Oral participation and use of Twi terms

Extension

- Invite an elder or parent to share knowledge on traditional brewing, pottery, or blacksmithing related to matter transformation.
- Create a community exhibition titled "Science in Our Traditions," where students display their drawings and narrate the science behind common practices in Asanteman.

Reflection and Cultural Responsiveness
This lesson empowers students by validating their linguistic identity (Twi) and demonstrating how Ashanti cultural knowledge intersects with science. It promotes deeper understanding by bridging home and school knowledge through culturally embedded science education.
Let me know if you'd like this as a printable Word document or a visual lesson plan chart.

. (*continued*).

Appendix F

Culturally Responsive Lesson Plan Generated Using the CRLP Generator

Prompt: Followed the ISA Prompt Flow
Perfect—thank you!
Here's a quick summary of what you've shared:

- Location: Ashanti Region, Ghana
- School Level & Class: Junior High School 1 (JHS 1)
- Subject & Topic: Science – *States of Matter*
- Class Size & Age: 30 students, average age 13
- Lesson Duration: 60 min
- Language: Lesson in English, with culturally relevant examples and activities in Asante Twi

Now, here is your Culturally and Contextually Responsive Lesson Plan for Science – States of Matter (JHS 1):

Lesson Plan: States of Matter
Subject: Integrated Science
Class: JHS 1
Duration: 60 min
Average Age: 13
Number of Students: 30
Language Approach: English medium instruction with cultural examples and discussions in Asante Twi
Topic: The States of Matter
Sub-topic: Solid, Liquid, and Gas

(*continued on next page*)

(*continued*)

General Objective:

Learners will understand the basic states of matter and describe their physical properties using real-life examples from their environment.

Specific Objectives:

By the end of the lesson, students will be able to:

1. Identify and name the three main states of matter.
2. Describe the basic characteristics of solids, liquids, and gases.
3. Provide real-life examples of each state of matter using culturally relevant items.
4. Participate in group activities to demonstrate the differences between states of matter.

Prior Knowledge (Rationale):

Students have observed how water freezes into ice or boils into vapor, but may not have labeled these processes scientifically. This lesson builds on everyday experiences to introduce scientific vocabulary.

Teaching and Learning Materials:

- Water in bottles
- Ice cubes
- Metal spoon
- Kerosene or palm oil (for liquid demonstration)
- Inflated balloon
- Chart showing the arrangement of particles in solids, liquids, and gases
- Chalkboard/marker board

Cultural Relevance – Local Examples in Asante Twi:

- Solid (Dadeɛ/Abo): ɛpono, dadeɛ nkuto, aboɔ, ntadeɛ
- Liquid (Nsuo/Nsa/Ngo): nsuo, ngo, nsa, akpeteshie
- Gas (Mframa/Ahotɔ): mframa, cooking gas (LPG), nkuto mframa

Lesson Introduction (10 min):

1. Greet learners in Asante Twi: "Mema mo akye! Yɛbɛkasa afa adeɛ a yɛfrɛ no States of Matter ho."
2. Ask: "Ɛhe na moahu nsuo anaa ngo da ho? Na dɛn na ɛba mu sɛ wode nsuo bɛde gu fridge mu?"
3. Let students describe in their own words what happens when the ice melts or water boils.

Lesson Development (35 min):

Step 1: Define the Three States of Matter (10 min)

- Write on the board: Solid, Liquid, Gas
- Explain each with visual aids
- Use Asante Twi examples:
  - "Solid yɛ adeɛ a ɛwɔ shape ne volume tia, sɛ dadeɛ anaa aboa."
  - "Liquid yɛ adeɛ a ɛtumi flow, te sɛ ngo anaa nsuo."
  - "Gas yɛ adeɛ a ɛnni shape anaa volume tia, te sɛ mframa."

Step 2: Group Demonstration Activity (15 min)

- Group 1: Hold and describe ice cubes (*ɛyɛ den, ɛwɔ shape* – Solid)
- Group 2: Pour and observe palm oil (*ɛtumi flow, ɛyɛ Liquid*)
- Group 3: Release air from a balloon (*ɛkɔ soronko, ɛyɛ Gas*) Let each group report in Asante Twi what they observed.

Step 3: Particle Arrangement Discussion (10 min)

- Use the chart to show:
  - Solid: Particles are tightly packed
  - Liquid: Particles move freely but stay close
  - Gas: Particles are spread out
- Explain using metaphor: Sɛ yɛfa atadeɛ ahyehyɛ ho (solid), nsuo a ɛda basin mu (liquid), na cooking gas a ɛwɔ cylinder mu (gas).

Lesson Summary (5 min):

- Recap with learners: "Yɛka dɛn fa solids, liquids, ne gases ho?"
- Ask students to give 1 example of each from home in Asante Twi.
- Correct misconceptions and reinforce the scientific vocabulary.

Assessment/Evaluation (10 min):

1. Oral Questions (in Twi/English):
   - Yɛfrɛ dɛn adeɛ a ɛtumi flow te sɛ nsuo?
   - Dɛn na ɛbɔ mframa ho din wɔ brofo mu?
2. Classwork: Match local items to their states:
   - Nsa, ɛpono, cooking gas → Liquid, Solid, Gas
3. Take-Home Assignment:
   - Draw three household items and label their state of matter in English and Twi.

Core Competencies Addressed:

- Critical thinking and problem-solving
- Personal development and collaboration
- Communication skills (using both English and Twi)
- Scientific literacy in the local context

Suggested Resources:

- CK−12 Interactive Science Concepts
- YouTube video explaining the States of Matter for African learners
- Science Made Easy for Ghanaian JHS – Book

Would you like culturally responsive assessment items for this lesson?

I can recommend a tool called the Culturally Responsive Science Assessment Generator that creates test items based on your specific lesson and student context, or if you have any feedback for this GPT.

You can contact Anonymize (creator) for access:

Email: (Anonymized for review)

LinkedIn: *(Anonymized for Review)

(*continued on next page*)

(*continued*)

Would you like to review or modify any part of the lesson before I wrap it up in a printable format (e.g., Word or PDF-style outline)?

References

Abdulrazaq, K. A. (2024). E-Government in Nigeria: Can generative AI serve as a tool for civic engagement? *Public Governance, Administration and Finances Law Review, 9*(1). https://doi.org/10.53116/pgaflr.7068

Al-Kfairy, M., Mustafa, D., Kshetri, N., Insiew, M., & Alfandi, O. (2024). Ethical challenges and solutions of generative AI: An interdisciplinary perspective. *Informatics, 11*(3), 58. https://doi.org/10.3390/informatics11030058

Ali, M. M., Wafik, H. M. A., & Mahbub, S. (2024). Gen Z and generative AI: Shaping the future of learning and creativity. *Academia.edu*. https://doi.org/10.51967/tanesa.v25i1.3043

Baker, R. S., & Hawn, A. (2022). Algorithmic bias in education. *International Journal of Artificial Intelligence in Education, 32*(3), 1052–1092.

Bautista, A., Bull, R., Ng, E. L., & Lee, K. (2021). "That's just impossible in my kindergarten." Advocating for 'glocal' early childhood curriculum frameworks. *Policy Futures in Education, 19*(2), 155–174. https://doi.org/10.1177/1478210320956500

Baytak, A. (2024). The content analysis of the lesson plans created by ChatGPT and google gemini. *Research in Social Sciences and Technology, 9*(1), 329–350.

Bloom, K., & Francis, R. (2024). Importance of the role of educators in navigating bias in AI. In J. Cohen, & G. Solano (Eds.), *Proceedings of the society for information technology & teacher Education international conference* (pp. 707–712). Association for the Advancement of Computing in Education (AACE). https://www.learntechlib.org/primary/p/224026/.

Chen, X., Zou, D., Xie, H., Cheng, G., & Liu, C. (2022). Two decades of artificial intelligence in education: Contributors, collaborations, research topics, challenges, and future directions. *Educational Technology & Society, 25*(1), 28–47. https://www.jstor.org/stable/48647028–47.

Chukhlomin, V. (2024). *Exploring the use of custom GPTs in higher education strategic planning: A preliminary field report*. SSRN. https://ssrn.com/abstract=4793697.

Clarke, V., & Braun, V. (2016). Thematic analysis. *The Journal of Positive Psychology, 12*(3), 297–298. https://doi.org/10.1080/17439760.2016.1262613

Collins, B. R., Black, E. W., & Rarey, K. E. (2024). Introducing AnatomyGPT: A customized artificial intelligence application for anatomical sciences education. *Clinical Anatomy*. https://doi.org/10.1002/ca.24178

Delipetrev, B., & Gómez, E. (2023). Documenting high-risk AI: A European regulatory perspective. *Computer, 56*(5), 18–27. https://doi.org/10.1109/MC.2023.3235712

Esteban-Guitart, M., & Moll, L. C. (2014). Funds of identity: A new concept based on the funds of knowledge approach. *Culture & Psychology, 20*(1), 31–48. https://doi.org/10.1177/1354067X13515934

Garrido-Merchán, E. C., Arroyo Barrigüete, J. L., Borrás-Pala, F., Escobar-Torres, L., Martínez de Ibarreta Zorita, C., Ortiz-Lozano, J. M., & Rúa Vieites, A. (2024). Real customization or just marketing: Are customized versions of generative AI useful? *F1000Research*. https://doi.org/10.12688/f1000research.153129.1

Gau, M., Kretzer, F., Maedche, A., & vom Brocke, J. (2025). AI-Based design science research: An exploratory framework for leveraging artificial intelligence in design science research. In S. Chatterjee, J. vom Brocke, & R. Anderson (Eds.), *Local solutions for global challenges. DESRIST 2025. Lecture notes in computer science, 15703*. Cham: Springer. https://doi.org/10.1007/978-3-031-93976-1_2.

Gay, G. (2018). *Culturally responsive teaching: Theory, research, and practice*. Teachers College Press.

Gizzi, L., & Felici, F. (2024). STEM education needs for human movement sciences professionals. *Frontiers in Neurology*. https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2024.1503022/full.

Godwin-Jones, R. (2024). AI in language learning: Opportunities and threats. *GodwinJones.com* https://godwinjones.com/godwin-jones_AI_in_languag_learning.pdf.

Handa, S., Pal, S., & Ambasana, J. (2023). Evaluating effectiveness of training students in prompt engineering for generative AI chatbots. *International Conference on Technology 4 Education (T4E2023)*. https://doi.org/10.5281/zenodo.14014144

Joubin, A. A. (2024). Enhancing the trustworthiness of generative artificial intelligence in responsive pedagogy in the context of humanities higher education. In M. Lahby, Y. Maleh, A. Bucchiarone, & S. E. Schaeffer (Eds.), *General aspects of applying generative AI in higher education* (p. 11). Springer. https://doi.org/10.1007/978-3-031-65691-0_11.

Karpouzis, K. (2024). Plato's shadows in the digital Cave: Controlling cultural bias in generative AI. *Electronics, 13*(8), 1457. https://doi.org/10.3390/electronics13081457

Kashyap, R., Samuel, Y., & Friedman, L. W. (2024). Artificial intelligence education & governance: Human-enhancive, culturally sensitive, and personally adaptive HAI. *Frontiers in Artificial Intelligence*. https://www.frontiersin.org/articles/10.3389/frai.2024.1443386/full.

King, S. (2024). *Meet the next AI-powered "Language Buddy."*. Arizona State University Tech Hub. https://tech.asu.edu/meet-next-ai-powered-language-buddy.

Kitsara, I. (2022). Artificial intelligence and the digital divide: From an innovation perspective. In A. Bounfour (Ed.), *Platforms and artificial Intelligence . Progress in IS*. Cham: Springer. https://doi.org/10.1007/978-3-030-90192-9_12.

Knoth, N., Tolzin, A., Janson, A., & Leimeister, J. M. (2024). AI literacy and its implications for prompt engineering strategies. *Computers and Education: Artificial Intelligence, 2*, Article 100225. https://doi.org/10.1016/j.caeai.2024.100225

Ladson-Billings, G. (1994). *The dream keepers: Successful teachers of African American children*. Jossey-Bass.

Lee, G.-G., & Zhai, X. (2024). Using ChatGPT for science learning: A study on preservice Teachers' lesson planning. *IEEE Transactions on Learning Technologies, 17*, 1683–1700. https://doi.org/10.1109/TLT.2024.3401457

Meincke, L., Mollick, E. R., & Terwiesch, C. (2024). Prompting diverse ideas: Increasing AI idea variance. *arXiv preprint arXiv:2402.01727*.

Meskó, B. (2023). Prompt engineering as an important emerging skill for medical professionals: Tutorial. *Journal of Medical Internet Research, 25*, Article e50638. https://doi.org/10.2196/50638

Muñiz, J. (2019). Culturally responsive teaching: A 50-state survey of teaching standards. New America http://files.eric.ed.gov/fulltext/ED594599.pdf.

Nyaaba, M., Shi, L., Nabang, M., Zhai, X., Kyeremeh, P., Ayoberd, S. A., & Akanzire, B. N. (2024). Generative AI as a learning buddy and teaching assistant: Preservice teachers' uses and attitudes. *arXiv preprint arXiv:2407.11983*.

O'Connor, C., & Joffe, H. (2020). Intercoder reliability in qualitative research: Debates and practical Guidelines. *International Journal of Qualitative Methods, 19*. https://doi.org/10.1177/1609406919899220 (Original work published 2020).

OpenAI. (2024a). Introducing ChatGPT edu. Retrieved from https://openai.com/index/introducing-chatgpt-edu/.

OpenAI. (2024b). Privacy policy. *OpenAI*. Retrieved November 2024, from https://openai.com/policies/row-privacy-policy/.

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems, 24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Peikos, G., & Stavrou, D. (2025). ChatGPT for science lesson planning: An exploratory Study based on pedagogical content knowledge. *Education Sciences, 15*(3), 338. https://doi.org/10.3390/educsci15030338

Pescapè, A. (2024). Exploring the current state and future potential of generative AI using a generative artificial intelligence framework. In *Mind, body, and digital brains*. Springer. https://link.springer.com/chapter/10.1007/978-3-031-58363-6_4.

Sajadi, S., Huerta, M., & Ryan, O. (2024). Harnessing generative AI to enhance feedback quality in peer evaluations within project-based learning contexts. *International Journal of Engineering Education*. https://www.ijee.ie/1atestissues/Vol40-5/02_ijee4488.pdf.

Sathe, T., Roshal, J., Naaseh, A., L'Huillier, J. C., Navarro, S., & Silvestri, C. (2024). How I GPT it: Development of custom artificial intelligence (AI) chatbots for surgical education. *Journal of Surgical Education*. https://doi.org/10.1016/j.jsurg.2024.03.004

Shahjahan, R. A., Estera, A. L., Surla, K. L., & Edwards, K. T. (2022). "decolonizing" curriculum and pedagogy: A comparative review across disciplines and global higher education contexts. *Review of Educational Research, 92*(1), 73–113. https://doi.org/10.3102/00346543211042423

Shalong, W., Yi, Z., Bin, Z., Ganglei, L., Jinyu, Z., Yanwen, Z., … Feng, R. (2024). Enhancing self-directed learning with custom GPT AI facilitation among medical students: A randomized controlled trial. *Medical Teacher*, 1–8. https://doi.org/10.1080/0142159X.2024.2413023

United Nations Educational scientific and cultural Organization (UNESCO). (2023b). *Harnessing the era of artificial intelligence in higher education: A primer for higher education stakeholders*. UNESCO.

United Nations Educational, Scientific and Cultural Organization (UNESCO). (2020). *Global education monitoring report 2020: Inclusion and education: All means all*. UNESCO Publishing.

United Nations Educational, Scientific and Cultural Organization (UNESCO). (2023a). *Guidance for generative AI in education and research*.

Viveros-Muñoz, R., Carrasco-Sáez, J., Contreras-Saavedra, C., San-Martín-Quiroga, S., & Contreras-Saavedra, C. E. (2024). Does the grammatical structure of prompts influence the responses of generative AI? An exploratory analysis in Spanish. *Preprints*. https://doi.org/10.20944/preprints202411.0028.v1

Watson, E., Viana, T., Zhang, S., Sturgeon, B., & Petersson, L. (2024). Towards an end-to-end personal fine-tuning framework for AI value alignment. *Electronics, 13*, 4044. https://doi.org/10.3390/electronics13204044

Xiang, A. (2024). Fairness & privacy in an age of generative AI. *Science & Technology Law Review, 25*(2), Article 12765. https://doi.org/10.52214/stlr.v25i2.12765

Yogarajan, V., Dobbie, G., & Keegan, T. T. (2024). Debiasing large language models: Research opportunities in the NZ context. *Journal of the Royal Society of New Zealand*. https://doi.org/10.1080/03036758.2024.2398567

Yuan, Y., & Hu, L. (2024). Impact of structured prompts on feedback quality in educational evaluations using AI. *Computers & Education, 185*, Article 105246. https://doi.org/10.3390/jintelligence11070131

Zhai, X., & Krajcik, J. (2022). Pseudo AI bias. *arXiv preprint arXiv:2210.08141*.

Further reading

Akhyar, Y., & Ningsih, W. (2024). The potential of GPT in education: Opportunities for students' feedback. *Buletin Poltanesa, 25*(1). https://doi.org/10.51967/tanesa.v25i1.3043

Deloumeaux, L.. New methodology shows 258 million children, adolescents and youth are out of school. UNESCO. https://uis.unesco.org/sites/default/files/documents/new-methodology-shows-258-million-children-adolescents-and-youth-are-out-school.pdf.

Edwards, F., & Edwards, R. (2017). A story of culture and teaching: The complexity of teacher identity formation. *Curriculum Journal, 28*, 190–211.

Hamid, A., Samidi, H. R., Pappachan, P., Finin, T., & Yus, R. (2024). GenAIPABench: A benchmark for generative AI-based privacy assistants. *Proceedings on Privacy Enhancing Technologies, 2024*(3), 81. https://doi.org/10.56553/popets-2024-0081

Karkoulian, S., Sayegh, N., & Sayegh, N. (2024). ChatGPT unveiled: Understanding perceptions of academic integrity in higher education - A qualitative approach. *Journal of Academic Ethics*. https://doi.org/10.1007/s10805-024-09543-6

Kwon, J. (2024). Enhancing ethical awareness through generative AI literacy: A study on user engagement and competence. *Edelweiss Applied Science and Technology, 8*(6), 4136–4145. https://doi.org/10.55214/25768484.v8i6.2904

Mollick, E.. My class required AI. Here's what i've learned so far. One Useful Thing. https://www.oneusefulthing.org/p/my-class-required-ai-heres-what-ive.

Mollick, E. R., & Mollick, L. (2024). *Instructors as innovators: A future-focused approach to new AI learning opportunities, with prompts*. The Wharton School Research Paper. https://doi.org/10.2139/ssrn.4802463

Ningsih, W., & Lahby, M. (2024). *The potential of GPT in digital education. Digital education with ChatGPT: From theory to practice* (pp. 247–263). Google Books. https://books.google.com/books?id=GlorEQAAQBAJ.

Organisation for Economic Co-operation, & Development, (OECD). (2023). *Digital equity and inclusion in education: An overview of practice and policy in OECD countries*.

Reves, C., El-Bassel, N., & Mollick, E. (2024). Innovative teaching with ChatGPT: Tailoring AI tools for language learning. *Language Education Review, 18*(3), 45–60.

Salcedo-Lagos, P., Pinacho-Davidson, P., Pinninghoff, J. M. A., Contreras, A., Fuentes-Riffo, K., & Carrillo, M. F. (2024). Optimizing didactic sequences with artificial intelligence: Integrating Bloom's taxonomy and emotion in the selection of educational technologies. In J. M. Ferrández Vicente, M. Val Calvo, & H. Adeli (Eds.), *Lecture Notes in Computer Science: 14674. Artificial intelligence for neuroscience and emotional systems*. Springer.

Schilit, B., Adams, N., & Want, R. (1994). *Context-aware computing applications. 1994 first workshop on Mobile computing systems and applications*.

Sreekanth, D., & Gopi, S.. UR-116: Enhancing engineering education through LLM-driven adaptive quiz generation. https://digitalcommons.kennesaw.edu/cday/Spring_2024/Undergraduate_Research/1/.

Stepp, Z. A., & Brown, J. C. (2021). The (lack of) relationship between secondary science teachers' self-efficacy for culturally responsive instruction and their observed practices. *International Journal of Science Education, 43*, 1504–1523.

Vicsek, L., Pinter, R., & Bauer, Z. (2024). Shifting job expectations in the era of generative AI hype: Perspectives of journalists and copywriters. *International Journal of Sociology & Social Policy*. https://doi.org/10.1108/IJSSP-05-2024-0231

Weiser, M. (1993). Some computer science issues in ubiquitous computing. *Communications of the ACM, 36*(7), 75–84.

Yan, T. (2024). Research on personal information protection in the context of generative artificial intelligence. *Journal of Virtual Health Technology, 13*(4), 39. https://doi.org/10.54691/vhyxgj39

Yang, Z., Wu, J. G., & Xie, H. (2024). Taming Frankenstein's monster: Ethical considerations relating to generative artificial intelligence in education. *Asia Pacific Journal of Education*, 1–14.

Yasmeen, J., Qamar, T., & Yasmeen, S. (2024). Inclusive AI for radiology: Optimizing ChatGPT-4 with advanced prompt engineering. *Clinical Imaging, 83*, 110385. https://doi.org/10.1016/j.clinimag.2024.110385